Unplugging the Artificial Intelligence Energy Emergency

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I. INTRODUCTION

This Article explores the assumptions underlying projections of extraordinary growth in artificial intelligence (AI)-driven data center energy demand and identifies the potential for reducing energy costs and demand through energy and environmental disclosures. The Article begins by examining the history of electricity demand over-

predictions and proposing research to address the over-prediction bias in assessments of future AI electricity demand. It then identifies potentially viable initiatives to shift the timing or amount of demand and eschews the tendency to propose government interventions on AI use or emissions that are infeasible for the foreseeable future. Focusing on household behavior, it identifies examples of successful disclosure programs and presents exploratory new empirical data to explain how electricity use and environmental disclosures may shift consumer behavior. Turning to corporate behavior, it then explores corporate environmental commitments and discusses ways to enhance their gap-filling. This Article concludes by identifying additional research initiatives to better understand and address the role of information in addressing the energy and environmental challenges arising from AI-driven electricity demand.

Projected electricity demand from AI data center growth has prompted claims that the United States is confronting an energy emergency,[1] rapid buildout of the natural gas supply and power generation infrastructure is necessary,[2] and corporations that have made net zero or other environmental pledges will face challenges meeting their commitments.[3]

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1 *See, e.g.*, Declaring a National Energy Emergency, 90 Fed. Reg. 8433 (Jan. 20, 2025). *But see* Editorial, *What "Energy Emergency?": The United States is Not Facing an Energy Crisis Today, But Trump Could Create One*, WASH. POST (Jan. 23, 2025), https://www.washingtonpost.com/opinions/2025/01/23/trump-energy-emergency/ [https://perma.cc/XEP7-6JFQ].

2 *See The Opportunities, Risks, and Challenges Associated with Growth in Demand for Electric Power in the United States*: *Hearing Before the Sen. Comm. Energy & Natural Res.*, 118th Cong. 8 (2024) (statement of Senator Barrasso) (claiming the U.S. needs "a reliable, steady, and balanced supply of electricity" that natural gas can provide); *id.* (statement of Ben Fowke, interim CEO and Pres. of American Electric Power) (requesting "significant investment in a diverse set of electric generation resources" including natural gas). *See also Report: Additional Natural Gas Infrastructure is Needed to Meet Energy Demand and Help Alleviate Increasing Energy Prices*, INTERSTATE NATURAL GAS ASS'N OF AM. (Sep. 11, 2023), https://ingaa.org/report-additional-natural-gas-infrastructure-is-needed-to-meet-energy-demand-and-help-alleviate-increasing-energy-prices/ [https://perma.cc/H4SB-G8AC].

3 *See* Lynne Kiesling, *Data Center Electricity Use IV: Unrealistic Emissions Targets?*, CENTER FOR

Institutional investors see this projected demand growth as "the beginning of a golden age" for investing trillions of dollars in energy infrastructure.[4] If the projected demand growth occurs, it would require not only large investments but also make it difficult to achieve energy-sector emissions reductions necessary to achieve climate goals.[5] Thus, projected demand growth poses challenges to the affordability of electricity and the ability to reduce environmental risks.

This Article offers new perspectives on three aspects of AI-driven data center electricity demand growth: (1) the accuracy of demand growth predictions; (2) the opportunity to reduce electricity use and environmental impacts from disclosure to users at the time of use; and (3) the gap-filling role of environmental commitments by AI companies. Often, proposed responses to AI-driven data center electricity demand simply accept demand projections and assume that it is not possible to shift the timing or amount of AI-driven electricity demand. They assume that the energy sector must satisfy demand increases in the projected amounts and that only efficiency gains with AI or government responses that increase the supply of electricity will address the problem. However, a growing body of social science, policy, and legal literature suggests that, in an era where

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TECH., SCI., & ENERGY (Aug. 22, 2024), https://ctse.aei.org/data-center-electricity-use-iv-unrealistic-emissions-targets/ [https://perma.cc/8ULR-BGM7] ("While tech companies are investing in renewable energy and efficiency improvements, the rapid expansion of data centers has still led to a notable increase in GHG emissions.").

4 *Regulators Weigh Billion-Dollar Utility Makeovers as Bills Rise for Consumers*, SCRIPPS NEWS (Sept. 27, 2025), https://www.scrippsnews.com/business/regulators-weigh-billion-dollar-utility-takeovers-as-bills-rise-for-consumers [https://perma.cc/UMA9-7GH9] (quoting BlackRock CEO Larry Fink saying "[t]here's a need for trillions of dollars investing in infrastructure related to our power grids, AI, the whole digitization of the economy," and that this represents "the beginning of a golden age" for investors).

5 *See* Alexandra Klass & Dave Owen, *Allocating Electricity*, 94 GEO. WASH. L. REV. 60, 64 n.17 (2026) (collecting sources); *see also* Michael P. Vandenbergh & Jonathan M. Gilligan, *Forks in the Road*, 31 DUKE ENV'T L. & POL'Y F. 163, 173 (2020) (concluding that decarbonization of the electric grid, electrification of transportation, and electrification of buildings could reduce U.S. carbon emissions by roughly 70%).

sufficient government action cannot be presumed, better theoretical work and more feasible responses to environmental problems will arise if scholars focus on demand-side impacts rather than mere federal and state options.[6] This Article suggests that each of the assumptions about AI-driven electricity demand should be examined, and that accepting them at face value may miss opportunities for viable public and private governance responses that will reduce energy costs and environmental risks.

II. ELECTRICITY DEMAND PROJECTIONS

Part I explores the reliability of the energy demand projections that are fueling the claims about an energy emergency by examining the history of demand projection, the role of data centers in future energy consumption, and economic and policy consequences of forecast uncertainty and utility regulation in energy infrastructure investment. As a much-repeated truism (often attributed to Mark Twain or Yogi Berra) states: predictions are hard, especially about the future.[7] The challenge with AI energy demand predictions is that they are often not merely wrong but tend to over-predict electricity demand. These over-predictions align with the financial incentives of the electric utility industry and the political incentives for governmental projections of economic growth. This stands in sharp contrast with the expectation that investment markets reflect the most reliable estimates of future demand. Markets are investing heavily with the presumption that the forecasts of

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6 *See, e.g.*, Matthew G. Burgess et al., *Supply, Demand, and Polarization Challenges Facing U.S. Climate Policies*, 14 NATURE CLIMATE CHANGE 134 (2024) (interdisciplinary social science author team advocating for feasibility assessment in climate policy); Kristian S. Nielsen et al., *Improving Climate Change Mitigation Analysis: A Framework for Examining Feasibility*, 3 ONE EARTH 325, 325–36 (2020); Michael P. Vandenbergh, *Environmental Law in a Polarized Era*, 38 J. LAND USE & ENV'T L. 51, 52–89 (2023) (advocating for feasibility assessment in climate governance).

7 Paul Douglas, *Have a Plan B for Rain This Weekend*, MINN. STAR TRIB. (Sep. 25, 2025, at 15:50 CT), https://www.startribune.com/douglas-have-a-plan-b-for-rain-this-weekend/601475520 [https://perma.cc/LD88-42SB].

sharp demand growth are correct.[8] The magnitude of these investments suggests a need for careful examination. We explore the tendency to over-predict demand below by examining the economics literature and legal literature and identifying research that can reduce the risk of this phenomenon with AI-data center electricity demand forecasts.

A. ECONOMIC LITERATURE: ELECTRICITY FORECASTS AND AI DATA CENTER GROWTH

Electricity demand forecasting plays a crucial role in energy infrastructure planning, guiding investments in power generation, transmission, and distribution. However, historical forecasts, whether from utility integrated resource plans (IRPs) or macroeconomic models such as the National Energy Modeling System (NEMS) used by the U.S. Energy Information Administration (EIA) are often subject to significant errors.[9] Understanding these inaccuracies and their economic implications is essential because energy demand will grow from the expansion of AI-driven data centers.

The literature investigating the retrospective accuracy of long-term energy demand projections is sparse. As noted in 2018 by Carvallo and colleagues,[10] few studies have conducted comparisons of demand projections and actual demand.[11] Carvallo et al. evaluated the forecast accuracies of 12 load serving entities (LSEs) across the Western Electricity Coordinating Council (WECC) for forecast years 2003–2011. Based on the

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8 *See* SCRIPPS NEWS, *supra* note 4.

9 *See* Juan Pablo Carvallo et al., *Long Term Load Forecasting Accuracy in Electric Utility Integrated Resource Planning*, 119 ENERGY 410, 410–22 (2018); Michael Wara, Danny Cullenward & Rachel Teitelbaum, *Peak Electricity and the Clean Power Plan*, 28 ELEC. J. 18, 20–22 (2015); Mark Dyson & Alex Engel, *The Billion-Dollar Costs of Forecasting Electricity Demand: How Distributed Energy Resources Can Ease the Challenges of Grid Investment Planning*, ROCKY MTN. INST. (Oct. 23, 2017), https://rmi.org/billion-dollar-costs-forecasting-electricity-demand/ [https://perma.cc/NQN5-YGLR].

10 Carvallo et al., *supra* note 9.

11 *See* H. L. Willis & J. E.D. Northcote-Green, *Comparison Tests of Fourteen Distribution Load Forecasting Methods*, PAS-103 IEEE TRANSACTS. POWER APPARATUS & SYS. 1190 (1984); Charles R. Nelson & Stephen C. Peck, *The NERC Fan: A Retrospective Analysis of the NERC Summary Forecasts*, 3 J. BUS. & ECON. STATS. 179, 179–87 (1985); BRIDGER M. MITCHELL, ROLLA EDWARD PARK & FRANCIS LABRUNE, PROJECTING DEMAND FOR ELECTRICITY: A SURVEY AND FORECAST (1986).

proportional errors provided by the study, the average error rate across the 12 LSEs was 7.58%.[12] For peak demand, the difference between the average annual growth rate for actual and forecast peak demand was much smaller, with overestimation within a couple of percentage points. Although not peer-reviewed, Dyson and Engel used demand forecasts reported to the Federal Energy Regulatory Commission (FERC) and found systematic overestimation of peak load demand for 2005–2015, observing a one-percentage-point overestimate for each year into the future.[13]

Carvallo et al. also identified the methods and variables used in each IRP issued by an LSE.[14] Most models relied on either time series or cross-sectional regressions, with input variables usually consisting of historical sales, cooling and heating degree days, population growth, and electricity prices. Some models included sector-specific variables (industrial/residential), and many LSEs relied on externally provided forecasts of demographic and economic inputs.

In addition, Carvallo et al. analyzed how LSEs evaluated the accuracy of their own predictions.[15] Among LSEs that reviewed past predictions, most attributed forecast errors to inaccurate projections of economic activity and demand-side management programs. The factors included in economic activity remain unclear, though, because gross domestic product (GDP) (national/regional) was only included in two IRP models. The 2013 Idaho Power Company IRP report suggested that the 2008 recession affected new residential

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12 *See* Carvallo et al., *supra* note 9, at 414 tbl.3. Each error (the difference between observed and projected electricity) was normalized by the total load and denoted as the proportional error.

13 *See* Dyson & Engel, *supra* note 9. Note that the average error ten years out was about ten percent and that the authors removed data from 2009-2010 to eliminate the effects of the Great Recession.

14 *See* Carvallo et al., *supra* note 9.

15 *Id*.

customer growth, implying that errors in population growth (and potentially electricity prices) contributed significantly to economic activity-related forecast errors.

Although not investigated by Carvallo et al., the inclusion of electricity prices in half of the models warrants discussion. Electricity prices are jointly determined by supply and demand, with consumer exposure limited by pass-through rates. Consequently, electricity prices are an endogenous variable, potentially introducing significant bias into prediction models. The EIA's Annual Energy Outlook (AEO) 2022 retrospective analysis revealed systematic underestimation of electricity prices in reports from 1998–2014.[16] Given a negative price elasticity of demand, consistent underestimation of future electricity prices likely leads to overestimates of electricity demand.

This observation aligns with analysis from Michael Wara of Stanford Law School. In a 2015 article with Danny Cullenward and Rachel Teitelbaum, Wara noted that the NEMS model systematically overestimated total electricity sales.[17] This is likely related to the fact that the NEMS model's macroeconomic module tends to significantly overestimate economic growth, which is a primary driver of energy demand growth.[18]

Population growth or declines and recession-related assumptions further complicate forecast interpretation. Future forecasting efforts should address emerging

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16 U.S. ENERGY INFORMATION ADMINISTRATION, TECHNICAL REPORT, ANNUAL ENERGY OUTLOOK 2022 RETROSPECTIVE: EVALUATION OF PREVIOUS REFERENCE CASE PROJECTIONS 21 (2022). Note that Annual Energy Outlook reports for 2015-2019 overestimated prices, followed by underestimation post-COVID-19.

17 *See* Wara, Cullenward & Teitelbaum, *supra* note 9, at 18.

18 *Compare* U.S. DEPARTMENT OF ENERGY, ANNUAL ENERGY OUTLOOK 2007 4 (2007) (predicting that inflation-adjusted US GDP would increase an average of 2.9% per year from 2005), *with* World Bank Group, *GDP (constant 2016 US$) – United States*, WORLD BANK GROUP: DATA, https://data.worldbank.org/indicator/NY.GDP.MKTP.KD?locations=US [https://perma.cc/RY2E-E8PG] (reporting that inflation-adjusted US GDP in 2019 was only 30 percent greater than in 2005). The year 2019 was taken for comparison to avoid artifacts caused by the 2020 COVID-19 pandemic.

factors such as technological advancements, distributed energy resources, electrification trends, and the importance of the timing of energy demand.[19]

As to AI-driven data center electricity demand, recent reports by Alex de Vries and the Electric Power Research Institute highlight significant potential increases in demand due to AI workloads, cloud computing, and cryptocurrency mining.[20] Although some IRPs now include data centers in their forecasts, existing models may struggle to incorporate rapidly changing demand patterns, including the following:

Load Shape Uncertainty: Unlike traditional loads, data centers exhibit nonlinear and highly variable demand.

Geographic Concentration: Clustering of data centers in specific regions can strain local grids.

Efficiency Gains: Advances in cooling technologies and AI optimization could reduce per-unit energy consumption but may not offset overall growth.

Overestimation of Market Demand for AI services: Firms are finding that adopting AI into their business processes may be less profitable than expected, so demand for AI services may fall short of projections.

Studies by Werner and Jarvis (2024) and Hill (2021) find evidence of

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[19] *See* Guadalupe Arcia-Garibaldi & Pedro Cruz-Romero, *Medium- and Long-Term Demand Estimation Integrating Emerging Technologies*, 230 ELEC. POWER SYS. RSCH. 110179, at 1 (2024); TYLER NORRIS ET AL. RETHINKING LOAD GROWTH: ASSESSING THE POTENTIAL FOR INTEGRATION OF LARGE FLEXIBLE LOADS IN US POWER SYSTEMS (2025) (discussing responses to AI electricity demand increases).

[20] *See* Alex de Vries, *The Growing Energy Footprint of Artificial Intelligence*, 7 JOULE 2191, 2191 (2023); ELEC. POWER RSCH. INST., POWERING INTELLIGENCE: ANALYZING ARTIFICIAL INTELLIGENCE AND DATA CENTER ENERGY CONSUMPTION 2 (2024). The International Energy Agency's (IEA) spring 2025 report projects that global data center electricity consumption will reach 945 Terawatt-hours (TWh) by 2030, representing approximately 3% of global electricity demand. This represents more than a doubling from the 2024 baseline of 415 TWh. The IEA projections suggest annual growth rates of approximately 15% for data centers—four times faster than other economic sectors—with particularly sharp increases in the United States and China. *See* INT'L ENERGY AGENCY, ENERGY AND AI 63 (2025).

overinvestment in capital projects in the electric utility industry, leading to significant welfare losses.[21] For instance, Hill examined electricity market restructuring in the late 1990s, finding that deregulation led to a 17% increase in capacity and associated overinvestment.[22] Similarly, Werner and Jarvis identified asymmetric adjustments in return on equity rates: capital cost increases were rapidly passed through, while decreases led to slower downward adjustments.[23] A 1% increase in the return on equity gap resulted in a three to four percent increase in capital assets, approximately $7 billion annually, with 80% of costs attributed to the electricity sector.

Recent studies find that 95% of companies that have adopted AI tools failed to realize measurable impacts on profits.[24] Moreover, much of the current demand for generative AI tools is driven by their purveyors pricing them well below the cost of delivering the services and absorbing multi-billion-dollar losses to build market share.[25] There are indications that as AI purveyors raise prices to break-even points, demand for AI

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21 *See* Karl Dunkle Werner & Stephen Jarvis, *Rate of Return Regulation Revisited* 1–3 (Energy Inst. at Haas, Working Paper 329, 2025), https://haas.berkeley.edu/energy-institute/research/abstracts/wp-329/ [https://perma.cc/E6HV-X3J2]; Alexander Hill, *Excessive Entry and Investment in Deregulated Markets: Evidence from the Electricity Sector*, 110 J. ENV'T ECONS. & MGMT. 102543, at 2 (2021).

22 *See* Hill, *supra* note 21, at 2.

23 *See* Werner & Jarvis, *supra* note 21, at 4.

24 Chris Cook, Clara Murray & Melissa Heikkilä, *America's Top Companies Keep Talking about AI — But Can't Explain the Upsides*, FIN. TIMES (Sep. 23, 2025), https://www.ft.com/content/e93e56df-dd9b-40c1-b77a-dba1ca01e473 [https://perma.cc/YV5C-E9EH]; Sheryl Estrada, *MIT Report: 95% of Generative AI Pilots at Companies Are failing*, FORTUNE (Aug. 18, 2025), https://fortune.com/2025/08/18/mit-report-95-percent-generative-ai-pilots-at-companies-failing-cfo/ [https://perma.cc/22GC-LW4W]; *see also* John Cassidy, *The A.I.-Profits Drought and the Lessons of History*, NEW YORKER (Aug. 25, 2025), https://www.newyorker.com/news/the-financial-page/the-ai-profits-drought-and-the-lessons-of-history [https://perma.cc/W5RB-V7TC] (discussing corporate disappointment with AI integration efforts and results).

25 Ed Zitron, *Anthropic Is Bleeding Out,* ED ZITRON'S WHERE'S YOUR ED AT (July 11, 2025), https://www.wheresyoured.at/anthropic-is-bleeding-out/ [https://perma.cc/A36V-SXSH] (reporting that "Anthropic is expected to lose $3 billion this year").

services may fall.[26] Further complicating matters, AI companies are encountering unexpected difficulties improving their products and reducing error rates. New generations of AI tools are proving more costly to train and less reliable than developers had hoped.[27] If these difficulties are not overcome, AI purveyors will have to raise prices even more, which could further suppress demand.

In sum, the economic literature suggests that energy demand forecasting remains a critical and complex tool for infrastructure planning. Historical inaccuracies in IRPs and macroeconomic forecasts, with a tendency to generate over-estimates of future demand growth, highlight the need for improved methodologies. Further, economic incentives and regulatory structures may exacerbate forecasting errors, leading to inefficient resource allocation. These difficulties are particularly important to consider given current concerns about AI-driven data center growth.

B. LEGAL LITERATURE: LAW AND POLICY IMPLICATIONS OF FORECASTS

As mentioned above, Wara, Cullenward, and Teitelbaum (2015) noted in the legal literature that the NEMS model frequently leads to overestimations of electricity sales and carbon emissions.[28] For example, as Figure 1 demonstrates, Wara et al. pointed out that the EIA's Annual Energy Outlook (AEO), which relies on the NEMS model, consistently

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[26] *Id.* (quoting users of AI-assisted programming tools, saying that recent price increases from Open AI make it unprofitable for startups to use those tools); *see also* Ed Zitron, *Anthropic and OpenAI Have Begun The Subprime AI Crisis,* ED ZITRON'S WHERE'S YOUR ED AT (July 7, 2025), https://www.wheresyoured.at/anthropic-and-openai-have-begun-the-subprime-ai-crisis/ [https://perma.cc/2TC5-TZFR].

[27] *See* Melissa Heikillä et al., *Is AI Hitting a Wall?*, FIN. TIMES (Aug. 15, 2025), https://www.ft.com/content/d01290c9-cc92-4c1f-bd70-ac332cd40f94 [https://perma.cc/E6ZH-GJ7A] (reporting that OpenAI's GPT version 5 "was the central icon of the entire approach of scaling to get to AGI , and it didn't work," quoting an expert saying, "the bubble burst. [The systems] were not making any money, and we were not finding enough high-value applications" to make AI businesses profitable).

[28] *See* Wara et al., *supra* note 9, at 20–22; Michael Wara, *Instrument Choice, Carbon Emissions, and Information*, 4 MICH. J. ENV'T & ADMIN. L. 261, 278 (2015).

overestimated retail electricity sales from the period between 1997 and 2013.[29]

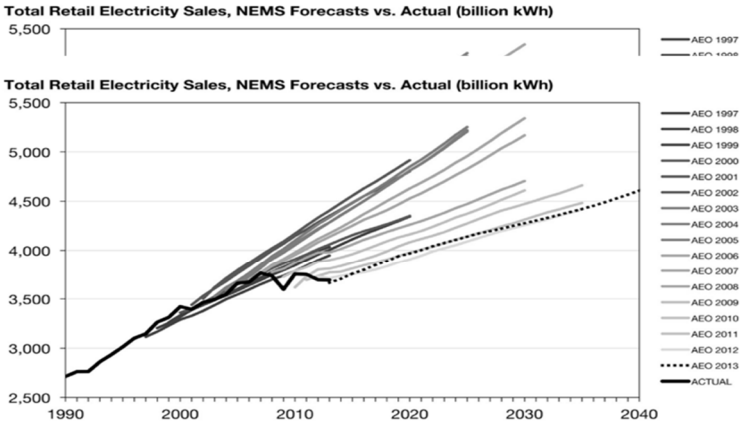


Figure 4: Although no model is perfect, NEMS has consistently overestimated total retail electricity sales since 1997. NEMS reference case forecasts for total retail electricity sales are shown here against the historical record, with AEO2013 highlighted as a dotted line.

Figure 1: "Total Retail Electricity Sales, NEMS Forecasts vs. Actual (billion kWh)[30]

In the context of predicting carbon emissions, AEO reference cases between 1999 and 2014 again demonstrated significant deviations between actual and projected emissions.[31] Multiple factors led to these deviations, including the Great Recession, shifts from coal to natural gas usage, and "a general decline in energy consumption per unit of GDP in the U.S. economy."[32]

Wara et al. argue that the "persistent bias and imprecision of the NEMS energy-

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29 The chart included as Figure 1 originally appeared in Wara et al., *supra* note 9 at 23. A caveat regarding the overestimation that Wara et al. identify is that all the papers they rely on include the Great Recession, so by definition all of the forecasts that occurred before 2008 for years after 2008 will likely be overestimated due to an exogenous shock, rather than a bias. The comparisons pre-2008 and 2010-onward suggest a smaller difference between observed and projected demand.

30 Wara et al., *supra* note 9, at 23.

31 Wara et al., *supra* note 28, at 276.

32 *Id.*

related CO2 forecasts" exists for two reasons.[33] First, the overprediction of fossil fuel usage leads to "more happy surprises[,]" producing under-predictions of prices while making energy crises easier to manage.[34] Second, NEMS is not apt to "simulate changes in the social and cultural factors that underlie and respond to economic and technological change."[35] The tendency of NEMS to overestimate GDP growth is likely a third important source of bias.[36]

The overprediction of energy demand also can be expected to induce greater capital spending by utilities on electric generation, and if utilities can convince the public utility commissions (PUCs) that the investments were prudent, they can recover their costs.[37] Thus, systemic over-prediction by utilities may produce greater revenues for the electric utility industry. In addition, retail and industrial customer demand and North American Electric Reliability Corporation reliability standards create strong incentives for utilities to provide a margin of safety to ensure that blackouts and brownouts do not occur.[38]

Policymakers and financial sector managers also have strong incentives to overestimate future economic growth. Policymakers benefit from presenting constituents

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33 *Id.* at 280.
34 *Id*. at 280–81.
35 *Id*. at 281.
36 *See* Wara et al., *supra* note 9.
37 For a discussion of the tendency of PUCs to approve utilities' requested cost recovery, *see* Lino Mendiola, *The Erosion of Traditional Ratemaking Through the Use of Special Rates, Riders, and Other Mechanisms*, 10 TEX. TECH ADMIN. L.J. 173, 174–76 (2008); Paul Joskow, *Regulatory Failure, Regulatory Reform, and Structural Change in the Electrical Power Industry*, BROOKINGS PAPERS ON ECON. ACTIVITY: MICROECONOMICS 125, 150–51 (1989) (discussing the factors driving significant reductions in the rate of growth in electricity demand during the 1970s, resulting in substantial excess generating capacity).
38 *See generally* NORTH AMERICAN ELECTRIC RELIABILITY CORPORATION, RELIABILITY STANDARDS FOR THE BULK ELECTRIC SYSTEMS OF NORTH AMERICA (2025), https://www.nerc.com/globalassets/standards/reliability-standards/rscompleteset.pdf [https://perma.cc/H9L6-BAWU].

with rosy forecasts of how their preferred policies will benefit the public.[39] Financial sector managers can more easily sell investment products to investors who believe the relevant sector will grow in value over time.

Past incentives for over-prediction regarding electric demand are even more acute regarding AI and data centers. The electric power industry has underestimated efficiency improvements on numerous occasions, even though efficiency has led to substantial reductions in electricity demand. For instance, the uptake of light-emitting diode (LED) lightbulbs contributed to the first downturn in per capita household electricity use in the United States since the second World War.[40] This is an important development because, as early as 2015, U.S. households emit as much carbon as all the sources from all the countries in South and Central America combined.[41]

The difficult question is whether the pattern of overestimation of electricity demand is also occurring for AI-driven data center use. Some analysts believe that the economic growth of the AI sector is vastly overestimated.[42] If, as they predict, the AI sector's value is overstated, underperformance of the sector could result in revisions to its energy demand. Of course, if perceived demand now facilitates rapid build-out of fossil-fuel-based electric generation, then by the time the extent of the overstated demand becomes clear, a gap will open between electric supply and demand. In turn, utilities might press PUCs to allow them

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39 *See* Jeffrey A. Frankel*, Over-optimism in Forecasts by Official Budget Agencies and Its Implications*, 27 OXFORD REV. ECON. POL'Y 536, 552 (2011); Jeffrey A. Frankel & Jesse Schreger, *Over-optimistic Official Forecasts and Fiscal Rules in the Eurozone*, 149 REV. WORLD ECONS 247, 271 (2013).

40 Lucas W. Davis, *Evidence of a Decline in Electricity Use by U.S. Households*, 37 ECON. BULL. 1098, 1101–02 (2017).

41 *See* Michael P. Vandenbergh & Jonathan M. Gilligan, *Beyond Gridlock*, 40 COLUM. ENV'T L.J. 217, 279 (2015).

42 *See, e.g.*, Jane Yoon, *AI Hype Has Echoes of the Telecom Boom and Bust,* FIN. TIMES (Feb. 13, 2024), https://www.ft.com/content/dc47c5f3-9bd4-4da0-a5cb-c795efd14c9c [https://perma.cc/9XN9-LDUT].

to recover those costs through continued operation of their plants at higher electricity rates for consumers. It also could create incentives for governments to back off carbon emissions initiatives such as carbon capture and storage that would otherwise increase the costs of the electric power supplied from an overbuilt system.

There is a long history of overestimating the energy needs of the internet, data centers, and cloud computing, which predates the challenge of generative AI. More than twenty years ago, a comprehensive analysis warned that at least one influential estimate of the internet's total power consumption was as much as eight times too high.[43] More recent analyses have found that as demand for data centers and cloud computing grew, the companies operating them invested heavily in energy efficiency innovations that reduced power demand for computing and ancillary needs, such as cooling. Consequently, the energy efficiency of data centers improved by an average of 20% per year between 2010 and 2020.[44]

For AI in particular, the example of the claimed efficiency gains from the DeepSeek model and new research on demand-side options for data center operations by Tyler Norris et al. raise questions about whether electricity demand predictions are once again

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43 Jonathan G. Koomey et al., *Sorry, Wrong Number: The Use and Misuse of Numerical Facts in Analysis and Media Reporting of Energy Issues*, 27 ANN. REV. ENV'T & RES. 119, 131 (2002), https://www.annualreviews.org/content/journals/10.1146/annurev.energy.27.122001.083458 [https://perma.cc/2VZ8-LXBF] (reporting that "the Huber and Mills estimate of internet power use was at least a factor of eight too high").

44 Eric Masanet et al., *Recalibrating Global Data Center Energy-use Estimates*, 367 SCI. 984, 985 (2020), https://science.sciencemag.org/content/367/6481/984 [https://perma.cc/HZ39-U7EM] (reporting that "the energy intensity of global data centers has decreased by 20% annually since 2010"); *see also* Jonathan Koomey & Eric Masanet, *Does Not Compute: Avoiding Pitfalls Assessing the Internet's Energy and Carbon Impacts*, 5 JOULE 1625, 1635–28 (2021), https://www.sciencedirect.com/science/article/pii/S2542435121002117 [https://perma.cc/6PJ6-VBLV]; JONATHAN KOOMEY, ZACHARY SCHMIDT, & TANYA DAS, ENERGY DEMAND GROWTH AND DATA CENTERS: A GUIDE FOR THE PERPLEXED (2025), https://bipartisanpolicy.org/download/?file=/wp-content/uploads/2025/02/BPC-Report-Electricity-Demand-Growth-and-Data-Centers-A-Guide-for-the-Perplexed.pdf [https://perma.cc/ZXH7-HANR].

underestimating efficiency gains.[45] DeepSeek reported that its model uses ten to forty times less energy than comparable models, although there are reasons to question these results.[46] This type of efficiency gains could substantially reduce demand. Alternatively, an extreme form of the Jevons Effect could arise,[47] in which more efficient energy use leads to lower prices, leading to greater demand. The net effect is unclear at this point, but electricity demand projections risk overestimating demand and stimulating costly and carbon-intensive utility investments if they do not account for substantial efficiency gains in AI and data centers. Moreover, training DeepSeek requires vast quantities of synthetic data derived from far more energy-hungry AI models, so it may not be possible for this technology to advance without the simultaneous development of far more energy-intensive models.[48]

Additionally, some analysts observe that the projected rise in demand for AI will be limited by economics as AI firms balance the profitability of selling AI services against

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45 *See* Christa Marshall, *'Game Changer?' What 'DeepSeek' AI Means for Electricity*, E&E NEWS: ENERGYWIRE, (Jan. 29, 2025), https://www.eenews.net/articles/game-changer-what-deepseek-ai-means-for-electricity/ [https://perma.cc/UG3S-CG3A]. DeepSeek was only able to achieve its energy-efficiency by using training data that came from OpenAI's energy-intensive system, and it may not be possible for energy-efficient AI to make progress without parasitically using the outputs of energy-intensive systems. *See, e.g.*, Lennart Heim, *The Rise of DeepSeek: What the Headlines Miss*, RAND RSCH. & COMMENT. (Jan. 28, 2025), https://www.rand.org/pubs/commentary/2025/01/the-rise-of-deepseek-what-the-headlines-miss.html [https://perma.cc/KMJ3-2DZL]. *See generally* NORRIS ET AL., supra note 19.

46 *See* Marshall, *supra* note 45. The DeepSeek energy use figures are highly disputed, and it appears as though Deep Seek was more effecient in the training phase, but not as much in the inference phase. *See* James O'Donnell, *DeepSeek Might Not Be Such Good News for Energy After All*, MIT TECH. REV. (Jan. 31, 2025), https://www.technologyreview.com/2025/01/31/1110776/deepseek-might-not-be-such-good-news-for-energy-after-all/ [https://perma.cc/UQ7W-6MKV].

47 *See* Joel Abrams, *DeepSeek Claims to have Cured AI's Environmental Headache. The Jevons Paradox Suggests It Might Make Things Worse*, CONVERSATION (Jan. 31, 2025), https://theconversation.com/deepseek-claims-to-have-cured-ais-environmental-headache-the-jevons-paradox-suggests-it-might-make-things-worse-248720.

48 *See, e.g.*, Heim, *supra* note 45.

the costs of electricity and computing hardware.[49] This consideration applies not only to the cost of inference (the cost of generating responses for users), but also to the cost of training models. Moreover, AI firms and researchers report that models are improving less and less with each generation, despite investing greater resources and energy in training them.[50] Daniel Castro concludes that "[a]s a result, many developers will likely focus more on optimizing their AI models rather than squeezing out ever-smaller improvements in accuracy because they will not receive a return on investment for building and operating larger models."[51]

Furthermore, if electricity demand rises quickly, competition from nuclear and renewables becomes less of a threat to natural gas and coal.[52] The natural gas industry not only has incentives to emphasize the speed and quantity of demand growth from AI and data centers, but also the need for "firm" power. Once pipelines and power plants are built, utilities will be less inclined to entertain competition from nuclear and renewables, and shutting down gas plants may meet resistance from employees, communities, and political

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49 DANIEL CASTRO, RETHINKING CONCERNS ABOUT AI'S ENERGY USE 8 (2024), https://-datainnovation.org/2024/01/rethinking-concerns-about-ais-energy-use/ [https://perma.cc/YJ7L-74U2] (observing that "[b]usinesses are profit-seeking enterprises and computing costs money; therefore, they are not going to offer services for long that cost more to operate than they receive in revenue. Either the energy costs for using AI will come down or how companies deploy AI will be limited by cost factors").

50 *Id.*; *See also*, Heikillä et al., *Is AI Hitting a Wall?*, *supra* note 27.

51 CASTRO, *supra* note 49.

52 *See* Adam Kay, *Fueling the AI Era: How Data Centers Rely on Natural Gas*, AM. GAS ASS'N: TRUE BLUE BLOG (May 28, 2024), https://www.aga.org/fueling-the-ai-era-how-data-centers-rely-on-natural-gas-2/ (arguing that "natural gas is the best tool currently available for helping data centers avoid potentially catastrophic power outages"); *see also* U.S. DEP'T OF ENERGY SEC'Y OF ENERGY ADVISORY BD., RECOMMENDATIONS ON POWERING ARTIFICIAL INTELLIGENCE AND DATA CENTER INFRASTRUCTURE 8 (2024), https://www.energy.gov/sites/default/files/2024-08/Powering%20AI%20and%20Data%20Center%20Infrastructure%20Recommendations%20July%202024.pdf [https://perma.cc/6EJU-HQVR] (finding that "[i]n most cases, [electricity providers, data center customers, and other large customers] see new natural gas capacity additions – in addition to solar, wind, and batteries – as the primary option available today to maintain reliability," implying that, if we build out now for projected electricity demand, natural gas is the primary energy source to use).

groups.[53] Regulatory limits imposed after these investments may also produce takings claims, either through legal challenges that raise constitutional concerns or within political debates.[54] Construing growing electricity demand as an emergency can garner public and defend government measures that overrides environmental constraints, whether through financial subsidies or regulatory changes.[55] Indeed, many gas and power companies are now reversing the old narrative that gas is a bridge to a clean energy economy, instead pointing to shortages of gas turbines to argue that clean energy should be seen as a bridge to a gas-dominated energy economy.[56]

C. RESEARCH AGENDA

The challenge for rational policymaking is to accurately predict electricity demand and to reduce demand over-prediction, which may be more dramatic for AI and data centers than for demand projections in earlier periods. To address these challenges, we suggest prompt, major research efforts that integrate uncertainty analysis and adopt more dynamic forecasting approaches. Research initiatives on several specific topics may also reduce the risk of errors in electricity demand projections:

Assess and Reduce Incentives for Over-optimistic Economic Growth Forecasts. The technology sector has a long history of optimistic predictions of future growth, leading

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53 *See* Christopher Serkin & Michael P. Vandenbergh, *Prospective Grandfathering: Anticipating the Energy Transition Problem*, 102 MINN. L. REV. 1019, 1019, 1022 (2018).

54 *Id*.

55 *See, e.g.*, Unleashing American Energy, 90 Fed. Reg. 8,353 (Jan. 20, 2025).

56 Stephen Stapczynski, Akshat Rathi & Josh Saul, *AI-Driven Demand for Gas Turbines Risks a New Energy Crunch*, BLOOMBERG (Oct. 1, 2025), https://www.bloomberg.com/features/2025-bottlenecks-gas-turbines/ [https://perma.cc/S6UD-5Z4Y] (reporting that "[g]as executives faced the ire of climate activists when they called gas a 'bridge' from coal to clean power. With the rapidly growing demand for power, these executives now think gas will be central to the future. . . . John Ketchum, chief executive officer of NextEra Energy Inc., the biggest US solar and wind developer, said clean energy should be a 'critical bridge' until gas plants can be built.").

to unrealistic valuations and subsequent bursting of bubbles. Similarly, long-term forecasts of GDP growth, both by governments and economists, show systematic overoptimism.[57] More realistic estimates of future economic growth at global, national, and sectoral scales, would lead to more realistic estimates of future energy demand.

Learn Lessons from Prior Technology Shifts. Research by energy scholars from several disciplines should identify the lessons for energy demand predictions and efficiency arising from the dotcom boom in the 1990s, cloud computing in the early 2010s, and cryptocurrencies in the late 2010s.

Pursue Energy Efficiency and Other Demand-Side Opportunities. Research by technology experts and engineers should explore options for energy efficiency at the data center level (such as power capping), the motivations for achieving efficiency, and the opportunities to shift the amount and timing of retail and corporate AI-driven energy demand.[58]

Address the Influence of Cost-Plus Utility Systems. The influence of cost-plus systems that PUCs use for determining utility compensation on predictions for future demand should be explored. Research on feasible modifications to rate structures enabling risks of over-prediction to be shared in efficient ways with shareholders and managers should also be studied.

Conduct a National Academies Consensus Study. To avoid biases in projections, a

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[57] *See* Masayuki Morikawa, *Uncertainty in Long-Term Macroeconomic Forecasts: Ex Post Evaluation of Forecasts by Economics Researchers*, 85 Q. REV. ECONS. & FIN. 8, 8–15 (2020).

[58] For example, Samsi et al. suggest power capping GPUs can be an effective tool for reducing inference energy. *See* SIDDHARTH SAMSI ET AL., FROM WORDS TO WATTS: BENCHMARKING THE ENERGY COSTS OF LARGE LANGUAGE MODEL INTERFERENCE 7–9 (2023). There could be work that investigates how power capping graphics processing units (GPUs) reduces the demand for inference (keeping in mind that ChatGPT processes 1 billion requests a day). These types of demand-side efficiency gains could be included in a study of electricity forecasts.

National Academies of Science, Engineering, and Medicine consensus study could be conducted on the lessons from prior predictions of electricity demand growth and the demand growth that is likely to arise from AI and data centers. The study should also explore ways to increase efficiency and ensure that demand growth does not disrupt the ability to maintain low energy burdens on electricity customers and meet environmental targets.

Improve Communication to Policymakers. Policymakers and the public need to be aware of the history of over-prediction and opportunities to shift demand when making decisions about future investments in power plants and transmission. Current projections may be accurate, but policymakers and the public should recognize the incentives to promote the higher demand forecasts to increase profits and discourage efforts to increase renewable and nuclear power.

III. CONSUMER ATTITUDES TOWARDS CARBON DISCLOSURE AND APPLICATIONS TO AI

Electricity use and environmental impact disclosure at the time of use may influence consumer behavior, encouraging consumers to use less electricity-intensive alternatives.[59] If search queries, search engines, or AI search firms differ in their energy use and environmental impacts, changes in user behavior could have substantial effects on the net electricity use and environmental impacts of AI and data centers. Employers that are cost-sensitive or who have made environmental commitments may have incentives to

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[59] *See* Zahra Rahmani Azad, Doron Cohen & Ulf J. J. Hahnel, *Recurrent Carbon Labels Induce Bipartisan Effects in Environmental Choices Under Risk,* 20 JUDGMENT & DECISION MAKING, at 1 (2024); Minghui Liu et al., *From Green Awareness to Green Behavior: The Impact of Information Disclosure Scenarios on Greener Shopping Channel Choices*, 16 SUSTAINABILITY 7944, 20 (2024). Recent research demonstrates that retail and corporate consumers may have difficulty assessing the differences in energy use among AI models. *See* Michael P. Vandenbergh, Ethan I. Thorpe & Jonathan M. Gilligan, *The Energy and Environmental Footprint of AI*, 15 MICH. J. ENV'T & ADMIN. L. (forthcoming 2026), https://papers.ssrn.com/sol3/papers.cfm?abstract_id=5400891 [https://perma.cc/ESM6-YTDA].

steer employees toward the least energy- and emissions-intensive employee behaviors. Additionally, those AI firms that have competitive advantages because of their lower energy use and other environmental impacts may have incentives to provide this type of disclosure to users even if they are not required by national and subnational governments.[60] Part II reviews the literature on information-driven behavior change, discusses the results of an exploratory study of disclosure options, and proposes more extensive research to explore the extent to which disclosures could shift the timing or amount of energy demand and thus reduce the costs and environmental impacts from AI-driven data center use.

Across a wide range of products, research demonstrates that easily understood disclosure at the point of decision can be effective when alternative options are available to users.[61] For instance, a study on consumer shopping choices comparing online and offline shopping revealed that both partial and full carbon emissions disclosure led more consumers to choose the less carbon intensive shopping avenue (in this case, online shopping).[62] However, responses to labels are complex and easily misunderstood. U.S. retail consumers are less interested in green labels than European retail consumers.[63] Research also suggests that although eco-labels on light bulbs can increase purchases by liberals, they can trigger identity effects that decrease purchases by conservatives,

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60 *See* SASHA LUCCIONI, YACINE JERNITE & EMMA STRUBELL, POWER HUNGRY PROCESSING: WATTS DRIVING THE COST OF AI DEPLOYMENT? 10–12 (2024) (finding significant energy use differences across general models). *See also* Liu et al., *supra* note 59, at 12; Jihoon Min et al., *Labeling Energy Cost on Light Bulbs Lowers Implicit Discount Rates*, 97 ECOLOGICAL ECON. 42, 42 (2014).

61 Khan M.R. Taufique et al., *Revisiting the Promise of Carbon Labelin*g, 12 NATURE CLIMATE CHANGE 132, 136 (2022).

62 Liu et al., *supra* note 59, at 12.

63 *See Insight of the Week: Consumer Purchasing Behaviors: EU vs USA*, GLOBESCAN (Dec. 18, 2020), https://globescan.com/2020/12/18/insight-of-the-week-consumer-purchasing-behaviors-eu-vs-usa/ [https://perma.cc/8GSN-8549].

regardless of efficiency.[64]

Additionally, it is a mistake to focus only on consumer responses when assessing the role of user-focused disclosure. For instance, research suggests that emissions reductions from carbon labeling may occur when companies identify efficiencies and attempt to reduce general brand risks from carbon disclosure, not just when consumers express a willingness to pay more for green goods.[65] The opportunity for time-of-use disclosure to reduce AI users' electricity demand and environmental impacts thus requires understanding (1) AI systems and the insights regarding the effects of disclosure on individual behavior, and (2) the effects of disclosure on corporate behavior, which may be only loosely tethered to direct consumer behavior.

A. THE STATE OF KNOWLEDGE ON ENERGY AND ENVIRONMENTAL LABELING

Individuals are quickly incorporating AI—specifically those based on large language models like those behind ChatGPT and Google Gemini—into their daily lives for tasks such as writing emails or searching the internet. As discussed above, the data centers necessary to run these generative AI models often require a large amount of electricity, and models' resource demands have led to significant carbon emissions increases by the various AI companies.[66] More critically, day-to-day use of AI requires substantial resource use—often much more than for computers and the internet to complete equivalent tasks. For example, one analysis found that using an AI-powered internet search tool used

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64 *See* Min *supra* note 60, at 48.

65 *See* Mark A. Cohen & Michael P. Vandenbergh, *The Potential Role of Carbon Labeling in a Green Economy*, 34 ENERGY ECON. S53, S56–S62 (2012).

66 Lauren Leffer, *The AI Boom Could Use a Shocking Amount of Electricity,* SCI. AM. (Oct. 13, 2023), https://www.scientificamerican.com/article/the-ai-boom-could-use-a-shocking-amount-of-electricity/ [https://perma.cc/24TT-MVUM]; INT'L TELECOMM. UNION & WORLD BENCHMARKING ALLIANCE, GREENING DIGITAL COMPANIES 42–43 (2025) [hereinafter GREENING DIGITAL COMPANIES].

approximately ten times the resources and, consequently, had ten times the impact of using a standard non-AI internet search tool.[67] AI-powered searches are now Google's default, suggesting that a traditional Google search is now many times more energy intensive than before the introduction of AI.

Although public awareness of AI's environmental risks is growing, people generally misestimate carbon emissions resulting from many of their individual behaviors and purchase decisions.[68] Further, the lack of transparency about AI energy use and carbon emissions suggests that an even higher percentage of people do not recognize or take into account AI's environmental impacts when deciding to use it.[69] And as AI tools are silently incorporated into a growing number of consumer products, consumers may often not have a choice about using AI. Additionally, less is known about how consumers react to the use

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67 Pranshu Verma & Shelly Tan, *A Bottle of Water Per Email: The Hidden Environmental Costs of Using AI Chatbots,* WASH. POST (Sept. 18, 2024), https://www.washingtonpost.com/technology/2024/09/18/energy-ai-use-electricity-water-data-centers/ [https://perma.cc/G3A3-XC8N]. The claim that ChatGPT uses 10 times more energy than a Google search has become a standard reference point in discussions about AI's environmental impact. Although this comparison is directionally accurate in suggesting AI's higher energy demands, the reality is more nuanced. The original estimates were based on extrapolations from older AI models and may not reflect current efficiencies or architectures. Google has disputed these figures, indicating their internal measurements show lower comparative energy usage, but they haven't released specific numbers. GEORGE KAMIYA & VLAD C. COROAMĂ, DATA CENTRE ENERGY USE: CRITICAL REVIEW OF MODELS AND RESULTS (2025), https://www.iea-4e.org/wp-content/uploads/2025/05/Data-Centre-Energy-Use-Critical-Review-of-Models-and-Results.pdf; *see* Amin Vahdat & Jeff Dean, *Measuring the Environmental Impact of AI Inference*, GOOGLE CLOUD: BLOG (Aug. 21, 2025), https://cloud.google.com/blog/products/infrastructure/measuring-the-environmental-impact-of-ai-inference/ [https://perma.cc/N35T-YFVB].

68 *See* Eric J. Johnson et al., *Widespread Misestimates of Greenhouse Gas Emissions Suggest Low Carbon Competence,* 14 NATURE CLIMATE CHANGE 707, 712–13 (2024); Karen P. Winterich, Rebecca W. Reczek & Tamar Makov, *How Lack of Knowledge on Emissions and Psychological Biases Deter Consumers from Taking Effective Action to Mitigate Climate Change,* 52 J. ACAD. MKTG. SCI. 1475, 1475–95 (2024).

69 Transparency may be growing, however, as demonstrated by Google's August 2025 release of new data on the energy use arising from a prompt. *See In a First, Google Has Released Data on How Much Energy an AI Prompt Uses*, MIT TECH. REV. (Aug. 8, 2025), https://www.technologyreview.com/2025/08/21/1122288/google-gemini-ai-energy/ [https://perma.cc/E4Q4-S4YT].

of carbon and other environmental labeling for services as opposed to goods.[70] To explore the potential for time-of-use disclosure to reduce the electricity demand and environmental impacts from AI-driven data center use, we suggest a series of behavioral studies that explore how disclosing electricity use and environmental impacts may influence users' willingness to opt in and opt out of using different types of AI models and use of different types of searches.

Psychology and economics research demonstrates that many people have difficulties understanding the environmental impacts associated with their everyday decisions, specifically regarding consumption. For instance, consumers often underestimate the energy used, and thus the emissions produced, by their various purchases, including food,[71] transportation,[72] and goods, such as clothing.[73] Environmental labeling—making a product's environmental impacts visible to consumers—allows for consumers to compare these impacts across equivalent goods, enabling them to make more sustainable purchasing decisions.[74] These labels come in many forms, each with different levels of information.[75]

Importantly, environmental labels enable consumers to make comparative judgments about the environmental impacts of either similar products with similar

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70 Eleni Fimereli & Susana Mourato, *Assessing the Effect of Energy Technology Labels on Preferences*, 2 J. ENV'T ECON. & POL'Y 245, 245–65 (2013).

71 Adrian R. Camilleri et al., *Consumers Underestimate the Emissions Associated with Food but are Aided by Labels*, 9 NATURE CLIMATE CHANGE 53, 53–58 (2018).

72 Shahzeen Z. Attari et al., *Public Perceptions of Energy Consumption and Savings*, 107 PROC. NAT'L ACAD. SCI. 16054, 16054–59 (2010).

73 Seth Wynes, Jiaying Zhao & Simon D. Donner, *How Well Do People Understand the Climate Impact of Individual Actions?*, 162 CLIMATIC CHANGE 1521, 1524 (2020).

74 Michael P. Vandenbergh, Thomas Dietz & Paul C. Stern, *Time to Try Carbon Labelling*, 1 NATURE CLIMATE CHANGE 4, 4–6 (2011).

75 For an overview of recent labeling initiatives, *see* Taufique, *supra* note 61, at 132–40.

functions (e.g., LED vs non-LED lightbulbs) or different products that fulfill similar functions (e.g., eating meat versus a vegetarian entrée). Because it is difficult for consumers to conceptualize carbon and other environmentally significant tradeoffs across various behaviors, these labels can help make their comparative judgments clearer.[76] Similar effects are predicted to arise regarding energy use.

Research on food consumption demonstrates that several types of labeling show promise in helping consumers make more sustainable decisions. For example, one technique, providing individualized feedback about carbon footprints through cell phones, showed an increase in consumers' sustainable decision-making during daily grocery shopping, specifically through a significant decrease in meat purchases.[77] However, not all carbon disclosure feedback needs to be individualized and resource-intensive to influence individual behavior. A less resource-intensive and lower-cost technique that shows promise regarding food sustainability is labeling food purchases with ordinal or traffic light-type messages (i.e., red vs. yellow vs. green light) regarding the level of emissions from a given meal choice.[78]

Although labeling can help consumers make more environmentally friendly purchasing decisions, little is known about how disclosing the environmental effects of new service-providing interactive technologies affects people's willingness to use them,

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[76] *See* Wynes, *supra* note 73.

[77] *See generally* Toke R. Fosgaard, Alice Pizzo & Sally Sadoff, *Do People Respond to the Climate Impact of Their Behavior? The Effect of Carbon Footprint Information on Grocery Purchases*, 87 ENV'T & RES. ECON. 1847 (2024).

[78] *See* Paul M. Lohmann et al., *Do Carbon Footprint Labels Promote Climatarian Diets? Evidence from a Large-scale Field Experimen*t, 114 J. ENV'T ECON. & MGMT. 102693 (2022); John Thøgersen & Kristian S. Nielsen, *A Better Carbon Footprint Label*, 125 J. CLEANER PROD. 86, 86–94 (2016); Florentine Brunner, et al., *Carbon Label at a University Restaurant–Label Implementation and Evaluation*, 146 ECOLOGICAL ECON. 658, 658–67 (2018); Magda Osman & Katie Thornton, *Traffic Light Labelling of Meals to Promote Sustainable Consumption and Healthy Eating*, 138 APPETITE 60, 60–71 (2019).

even when these technologies are useful in day-to-day household and workplace tasks. Important questions include the following: Do people alter their type, timing, or usage of a service when presented with a disclosure about its environmental impacts? How does the service and its specific tasks affect how people will react to the label? Are some AI firms motivated to provide disclosure to their customers? Are employers motivated to provide disclosure to their employees? Lastly, are less environmentally impactful alternatives easily available to consumers who want them?[79]

Due to the lack of definite answers to these questions, standard retail users may not be the most promising sources of information. The energy impact of querying generative AI at an individual level was negligible up to this point compared to other activities, such as a Zoom call. However, newer models and growth in AI usage may increase concerns regarding individuals' energy usage. For instance, generative AI has moved beyond chatbots to more computationally intensive deployments. Agentic AI seeks to complete a task autonomously on behalf of a user, and such agentic systems have become a central direction of new development.[80] Reasoning functionality—popularized by DeepSeek and now standard across major models—instructs a model to break queries into a series of steps and consider each portion of its answer before responding.[81] In doing so, the model executed many additional functions that were computationally intense but hard to measure because they occurred without user awareness (e.g., a recent query induced OpenAI's deep

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79 *See* Paul C. Stern et al., *Design Principles for Carbon Emissions Reduction Programs*, 44 ENV'T. SCI. & TECH. 4847–48 (2010), http://dx.doi.org/10.1021/es100896p (noting that interventions to reduce individual and household carbon emissions must make it convenient and simple for people to choose low-carbon options).

80 This development raises questions such as: What standards or protocols, if any, should be in place for this agentic activity? How is that activity measured and reported?

81 DEEPSEEK AI, DEEPSEEK-R1: INCENTIVIZING REASONING CAPABILITY IN LLMS VIA REINFORCEMENT LEARNING 2 (2025), https://arxiv.org/pdf/2501.12948.

research to "think" for 18 minutes before answering).[82] These developments suggest that, in addition to the energy and environmental impacts of standard retail users, research should examine how energy use and environmental information influence application developers who build software on top of large language models and make strategic design decisions about which models to use and when. This tradeoff is often thought of in monetary terms, but energy and environmental concerns could also play a role.[83]

B. PRELIMINARY STUDY

To examine the propensity for people to alter their day-to-day use of AI upon being informed about its environmental impacts, we conducted a preliminary survey-based empirical study. We collected a nationally representative sample of 300 participants,[84] and surveyed the participants about their likelihood of using AI for different daily tasks both before and after they read a paragraph about the amount of resources necessary for AI to generate responses.[85] Notably, 80% of participants reported using AI for various day-to-day tasks at least once a month, with 20% reporting using AI at least once a day.

The results of this preliminary study suggest that disclosure may be a promising option for reducing the electricity demand (and thus, the cost) and environmental impacts associated with AI-driven data center use. We found that disclosing AI's environmental

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82 *See* INT'L ENERGY AGENCY, *supra* note 20, at 43.

83 *See Frugal AI Challenge,* FRUGAL AI CHALLENGE, https://-frugalaichallenge.org/ [https://perma.cc/UV8X-FWTB].

84 This sample had nationally representative quotas for gender, age, race, and political party. All data collection procedures were approved through Vanderbilt University's Institutional Review Board (Approval #240102).

85 The paragraph presented to participants used statistics about electricity and water consumption from this Washington Post article: Pranshu Verma & Shelly Tan, *A Bottle of Water Per Email: The Hidden Environmental Costs of Using AI Chatbots*, WASH. POST (Sept. 18, 2024), https://www.washingtonpost.com/technology/2024/09/18/energy-ai-use-electricity-water-data-centers/ [https://perma.cc/C3QU-LEC9].

impacts made people feel more negatively about using AI. More importantly, environmental resource implications induced people to indicate that they were less likely to use AI in the future for daily tasks. Specifically, after reading the disclosure, people indicated they would be less likely to use AI for generating writing, helping with school or work, searching the internet, and entertainment. We did not explore the extent to which others would switch the forms of AI they used, providers, or other aspects of AI use that could affect energy demand. Additionally, the preliminary research only surveyed behavioral intentions, and more research must be done to investigate if this finding will generalize to actual behavior.

C. Additional Research

The preliminary study's results suggest that additional empirical research on the disclosure of AI-driven electricity use and environmental impacts would be valuable. We propose several lines of studies meant to investigate people's understanding and behavioral responses to disclosures about the environmental impacts of AI and the feasibility of providing such information to users. To accomplish this, the work should have three main goals.

First, the research should seek to understand the extent to which people associate AI with significant energy use and other environmental impacts. Do people recognize the environmental impacts of the services that AI provides? Does providing people with new information about the environmental impacts of AI cause them to change their beliefs? Are people influenced by motivated reasoning or confirmation bias when processing this information?

Second, the research should test the extent to which providing disclosures about the energy use or environmental impacts of AI influences people's willingness to opt in or

out of using AI in a behavioral setting, choose different AI tools, or make other behavioral adjustments. This research should also test how these effects might differ after implementing the use of environmental labels on purchasing goods.[86] It should also examine the different boundary conditions regarding the tasks and the influence of disclosures.

Third, the research should seek to use the information garnered from the first two goals to develop the most influential disclosure for reducing the energy use and environmental impacts arising from AI. Even though carbon labeling and other forms of environmental disclosure show promise in increasing consumers' sustainable decision-making, the characteristics of a disclosure that would induce more sustainable decisions regarding AI use remain unclear. In general, AI firms can be expected to have a financial interest in reducing electricity consumption to the extent electricity costs dominate their marginal costs and, thus, their operating profit margins. If so, getting users to simplify or optimize their queries for the same output can be expected to increase their profits (holding all else fixed). An important approach is to study whether this expectation holds true in practice and to assess the range of incentives that AI firms, data center firms, and employers may have to reduce AI's electricity usage and environmental impacts. This assessment should clarify which organizations are likely to have incentives to adopt disclosure

[86] We recognize that a great deal of heterogeneity exists in AI (e.g. text classification, extractive question answering, token classification, object detection, multitask extractive question answering, text generation, summarization, and multitask summarization). Some tasks do not have an appropriate substitute (like a normal Google search), other than doing something by hand. Thus, the behavioral responses will be conditional on the type of task. *See* Luccioni et al., *supra* note 60 (finding significant differences in energy use across general models). Therefore, given the value of using AI for a given task, it might be useful to test how disclosure moves people from using general AI tools, to more specialized AI for the given task. The goal might be to induce users to substitute the use of low-energy tools for high-energy ones, a likely more externally valid approach than using general AI less (or not at all) for a given task. *See id.*

systems.[87]

D. BROADER IMPACTS

Understanding individuals' reactions to the environmental impacts of AI might have significant implications beyond reducing the energy use and environmental impacts arising from using AI. These results may also provide insights that could affect public and private policymakers' ability to affect corporate decision making about AI design choices and data center electricity use. For example, AI providers that have a competitive advantage on energy efficiency or environmental impacts may have incentives to adopt disclosure systems, and employees that use AI may have incentives to use those AI providers. These incentives will increase to the extent corporations, state and local governments, civic and cultural organizations, universities, and other employers have cost constraints or make environmental supply chain commitments and disclose the environmental impacts arising from AI use.

Similarly, if AI providers know that people may choose to opt out of AI-generated internet searches or shift the timing or type of use after learning about their product's energy use and environmental impacts, they may have additional incentives to tailor their future models to include sustainable options, such as an opt-out ability, identifying efficiencies in their AI systems and data centers, using more efficient data centers, making or complying with commitments to reduce the environmental impacts from AI-driven data

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[87] For instance, competition in the AI sector implies that many firms will likely move to reduce training (and perhaps even inference) costs. This includes both electricity and hardware optimization. GenAI's over $100 million investment compared to DeepSeek's approximately $6 million investment implies large potential savings for large US firms optimizing their training, regardless of the climate implications. However, the actual size of DeepSeek's investment remains a topic of debate. *See, e.g.*, *The Geopolitics of DeepSeek: Narratives, Perception, and the AI Race,* THE SOUFAN CTR: INTELBRIEF (Feb. 6, 2025), https://thesoufancenter.org/intelbrief-2025-february-6/ [https://perma.cc/2Z8D-7G7Q].

center electricity use, and supporting public and private governance initiatives that facilitate these objectives. Knowing how people react to discovering the environmental impacts of a widely adopted technology, such as ChatGPT, also has implications for understanding how to communicate about and mitigate these impacts for future technologies.

IV. CORPORATE CARBON COMMITMENTS BY AI FIRMS

A 2025 Morgan Stanley analysis suggests that the electricity demand from data centers serving AI will increase their carbon emissions through 2030 by roughly 2.5 billion tons.[88] For context, Japan's annual emissions are roughly one billion tons.[89] Thus, the 2.5 billion tons of AI-driven data center emissions by 2030 will increase global emissions by a few countries worth, complicating the already difficult task of developing a global effort to meet emissions targets.

The next question must then be: What can governments do to reduce the demand-based stress on electricity generation and lower carbon emissions? Pressure will arise from the European Union (EU) and some states, but the U.S. federal government and many states are unlikely to promote decarbonization policies for at least the next several years.[90] Roughly half of the states also are likely to oppose climate policies for the foreseeable

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88 Laila Kearney, *Global Data Center Industry to Emit 2.5 Billion Tons of CO2 through 2030, Morgan Stanley says*, REUTERS (Sept. 3, 2024), https://www.reuters.com/markets/carbon/global-data-center-industry-emit-25-billion-tons-co2-through-2030-morgan-stanley-2024-09-03/ [https://perma.cc/K5DW-48M4].

89 M. CRIPPA ET AL., GHG EMISSIONS OF ALL WORLD COUNTRIES 11 (2025).

90 *See*, *e.g.*, Matt McGrath, *Trump Revokes Landmark Ruling that Greenhouse Gases Endanger Public Health*, BBC (Feb. 13, 2026), https://www.bbc.com/news/articles/cn0zdd7yl4vo [https://perma.cc/ZS9C-YTAF].

future.[91] We suggest that the better question is: What can any institution do to reduce the demand-based stress on electricity generation, costs, and environmental impacts?[92] We explore the potential for private environmental governance responses below.

A. PUBLIC GOVERNANCE

AI firms, data centers, and utilities will face critical investment decisions over the next several years. The federal government is highly unlikely to pursue decarbonization strategies, and the Supreme Court has signaled that it will reject broad exercises of the U.S. Environmental Protection Agency's (EPA) current statutory authority to address new environmental problems.[93] In fact, 2025 executive orders on the energy emergency are a signal that the federal government will pursue laws, policies, and programs discouraging rather than promoting the energy transition.[94]

State governments could motivate emissions reductions from AI-driven data center electricity use, and roughly half of the states have laws or policies designed to reduce carbon emissions. In addition, recent Vermont and New York statutes modelled on the federal Superfund statute may have attached financial risks to fossil fuel emissions.[95] State common law nuisance and consumer deception actions have also created limited incentives

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91 *See, e.g.*, *id.*; Spencer Grubbs, *Fighting A Fossil Fuels Boycott*, TEX. COMPTROLLER: FISCALNOTES (May 2023), https://comptroller.texas.gov/economy/fiscal-notes/archive/2023/may/fossil-fuels/ [https://perma.cc/2LDT-3YNC] (describing Texas State Government efforts to curtail corporate ESG efforts).

92 *See* Vandenbergh, *supra* note 6, at 75–76.

93 *See* McGrath, *supra* note 90 West Virginia v. EPA, 597 U.S. 697 (2022); Loper Bright Enterprises v. Raimondo, 603 U.S. 369 (2024).

94 *See, e.g.*, Michael Gerrard, *Trump Executive Orders on Energy and Environment: A Quick Summary*, COLUM. L. SCHOOL, CLIMATE LAW: SABIN CENTER BLOG (Jan. 21, 2025), https://blogs.law.columbia.edu/climatechange/2025/01/21/trump-executive-orders-on-energy-and-environment-a-quick-summary/ [https://perma.cc/Y67H-MGUP]. For a discussion of longer-term impediments to adoption of pollution control statutes, *see* Vandenbergh, *supra* note 74 at 52-89.

95 VT. STAT. ANN. TIT. 10, § 597 (2024); N.Y. ENV'T CONSERV. LAW § 76-0101 (McKinney 2024); Chamber of Com. of the U.S. v. Moore, No. 2:24-cv-01513 (D. Vt. filed Dec. 30, 2024).

for carbon emissions reductions.[96]

Some AI firms, data centers, and power plants are located in states that create regulatory pressure for carbon emissions reductions, but many are not (and likely will not be) located in those states.[97] Roughly half of the states rejected carbon emissions reductions requirements or actively litigated against climate regulations.[98] Similarly, the effects of state tort liability are unclear. There is a risk of a new data center "race-to-the-bottom" in which new data centers and the power plants that serve them have incentives to locate in states that do not regulate emissions or create statutory or common law tort liability risks.[99]

Pressure for energy use or carbon emission reductions could also arise from foreign governments and international governmental organizations. For instance, China already took steps to regulate AI-driven data center use.[100] The EU AI Act includes measures

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96 *See, e.g.*, Am. Elec. Power Co. v. Connecticut, 564 U.S. 410 (2011) (concluding that the authority to regulate greenhouse gases in the Clean Air Act displaces tort law public nuisance actions).

97 Joshua A. Geltzer et al., *State Regulation of Data Centers: Emerging Trends and Potential Legal Complexities*, WILMERHALE CLIENT ALERT (Feb. 23, 2026), https://www.wilmerhale.com/en/insights/client-alerts/20260223-state-regulation-of-data-centers-emerging-trends-and-potential-legal-complexities [https://perma.cc/ENB8-5UGX]; Ruby Wincele, *Research Brief: Identifying Priority States for Data Center Regulation*, CLIMATE XCHANGE (Feb. 12, 2026), https://climate-xchange.org/2026/02/research-brief-identifying-priority-states-for-data-center-regulation/ [https://perma.cc/S5BD-6344].

98 *See* Hilary Howard, *22 States Sue to Block New York Law Targeting Fossil Fuel Companies*, N.Y. TIMES (Feb. 6, 2025), https://www.nytimes.com/2025/02/06/nyregion/climate-change-superfund-act-lawsuit.html [https://perma.cc/8FKK-3WZL]; *State Climate Policy Maps*, C2ES, https://www.c2es.org/content/state-climate-policy/ [https://perma.cc/FL6F-2AHE] (last visited Mar. 8, 2026).

99 For a discussion of the race to the bottom concept, *see* Richard L. Revesz, *Rehabilitating Interstate Competition: Rethinking the "Race-to-the-Bottom" Rationale for Federal Environmental Regulation,* 67 N.Y.U. L. REV. 1210 (1992), and Kirsten Engel, *State Environmental Standard-Setting: Is There a "Race" and is it "to the Bottom"?*, 48 HASTINGS L. J. 271 (1997).

100 Chinese authorities have begun restricting the expansion of energy-hungry data centers in certain regions. *See* Kenza Bryan, *Data Centers Curbed as Pressure Grows on Electricity Grids*, FIN. TIMES (Feb. 12, 2024), https://www.ft.com/content/53accefd-eca7-47f2-a51e-c32f3ab51ad5 [https://perma.cc/E46Q-XPKK]. In addition, Chinese cities have introduced rules requiring new data centers (which include AI computing centers) to meet aggressive efficiency benchmarks. For instance, major Chinese data hubs must achieve a Power Usage Effectiveness (PUE) around 1.3 or lower for new facilities. *See* Fiona Brocklehurst, Policy Development on Energy Efficiency of Data Centers, INT'L

addressing the environmental impact of AI systems.[101] Although its primary focus is on AI risks to safety and rights, the final legislation includes provisions on energy and resource efficiency, albeit more limited than initially proposed.[102]

The AI Act's recitals explicitly recognize climate and environmental protection as public interests to be safeguarded,[103] signaling that AI's carbon footprint is an EU policy concern. The most concrete requirement in the final AI Act applies to providers of general-purpose AI models (large foundation models, such as GPT-5 or Claude). Under Article 53[104] and Annex XI,[105] providers of these models must prepare technical documentation, including "known or estimated energy consumption of the model." When actual energy consumption is unknown, providers may base estimates on computational resources used during training.[106] This requirement should mean that providers document the electricity and computing resources consumed during model development. Providers must make this documentation available to the European Commission's AI Office and national competent

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ENERGY AGENCY (Feb. 2024), https://www.iea-4e.org/wp-content/uploads/2024/02/Policy-development-on-energy-efficiency-of-data-centres-draft-final-report-v1.05.pdf#:~:text=Medium%20ambition%3A%20maximum%20PUE%201,data%20centcen%20or%20for [https://perma.cc/T9ZF-5PT8]. China has also set an objective that about 25% of data center energy should come from non-fossil sources by 2030, pushing AI cloud providers toward solar, wind, and hydro power. *See* Matthew Connatser, *Small Data Centers Face the Axe Under China's New Energy Policy,* THE REGISTER (Aug. 13, 2024), https://www.theregister.com/2024/08/13/china_green_policy_tech_elements/#:~:text=polpol%20www,wind%2C%20hydro%2C%20solar%20and [https://perma.cc/W6WC-RQT7].

101 Council Regulation 2024/1689, Artificial Intelligence Act, 2024 O.J. (L 1689) 1, Annex XI [hereinafter EU AI Act]; *id.* at art. 40.

102 *See* José Renato Laranjeira de Pereira, *The EU AI Act and Environmental Protection: The Case for a Missed Opportunity*, HEINRICH BÖLL STIFTUNG (Apr. 8 2024), https://eu.boell.org/en/2024/04/08/eu-ai-act-missed-opportunity.

103 *See* EU AI Act, *supra* note 101, at recitals 1, 2, 8, art. 1(1).

104 *Id.* at art. 53.

105 *Id.* at *Annex XI: Technical Documentation Referred to in Article 53(1), Point (a) – Technical Documentation for Providers of General-Purpose AI Models | EU Artificial Intelligence Act*, https://artificialintelligenceact.eu/annex/11/ (last visited Oct. 9, 2025).

106 *Id.* at Annex XI, § 1, ¶ 2(e).

authorities upon request.[107] These obligations for general-purpose AI models became applicable on August 2, 2025, although models released before that date have until August 2, 2027 to comply.[108] The AI Office has since operationalized this requirement through a standardized Model Documentation Form, which calls for training and inference energy use to be reported in megawatt-hours—though that information is shared only with the AI Office and national regulators upon request, not with downstream developers or the public.[109] Similarly, the Organization for Economic Co-operation and Development (OECD) AI Principles (adopted by over forty countries in 2019, including the US and EU members) affirm that AI should drive inclusive growth and sustainable development.[110]

B. PRIVATE GOVERNANCE

Governments are not the only source of motivation to reduce carbon emissions from AI and data centers. Private environmental governance initiatives are also a potential source of pressure for reduced electricity demand and lower environmental impacts.[111] For instance, in July 2025, an international standard-setting organization that includes private and public organizations, the International Organization for Standardization (ISO) and the

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[107] *Id.* at art. 53(1)(a).

[108] *Id*. at art. 113(b); *id.* at art. 111(3).

[109] EUR. UNION, EUR. COMM'N, *The General-Purpose AI Code of Practice* (Apr. 23, 2026), https://digital-strategy.ec.europa.eu/en/policies/contents-code-gpai [https://perma.cc/RU32-WNAD].

[110] *See OECD AI Principles Overview,* OECD AI POL'Y OBSERVATORY, https://oecd.ai/en/ai-principles [https://perma.cc/63VQ-RNNG] (last visited Mar. 8, 2026). The language is broad but states that AI's benefits should not come at the cost of environmental harm. In 2024, the OECD updated these principles to explicitly reference environmental sustainability as a key consideration for AI governance. *See* Kathrin Gardhouse, *Updated OECD AI Principles to Keep Up with Novel and Increased Risks from General Purpose and Generative AI*, LIMINA (June 12, 2024), https://www.getlimina.ai/en/blog/oecd-ai-principles-2024 [https://perma.cc/6BAR-789V].

[111] *See, e.g.*, *AI Energy Score Leaderboard*, HUGGING FACE, https://huggingface.co/spaces/AIEnergyScore/Leaderboard [https://perma.cc/X7WQ-AMQH] (last visited Mar. 8, 2026); *Salesforce Joins Technology and Academic Leaders to Unveil AI Energy Score Measuring AI Model Efficiency*, SALESFORCE: NEWS & INSIGHTS (Feb. 10, 2025), https://www.salesforce.com/news/stories/ai-energy-score/ [https://perma.cc/87LX-ZTJT].

International Electrotechnical Commission (IEC) joint technical committee on AI, published ISO/IEC Technical Report (TR) 20226:2025, "Environmental sustainability aspects of AI systems," the first international standard specifically addressing AI's environmental impacts.[112] The standard establishes a framework for assessing environmental impacts across the AI system lifecycle and defines metrics for energy efficiency, water usage, carbon emissions, and other environmental categories. Whether this framework will be widely adopted by AI providers is unknown, but it represents an important step toward standardizing how the industry measures AI's environmental footprint.

Additionally, responding to pressure from advocacy groups, employees, retail customers, investors, insurers, and lenders, many technology companies have joined climate mitigation initiatives, made net zero commitments, and sourced a substantial share of their electricity from renewables.[113] The technology sector also has the earliest average target year among major economic sectors for achieving net zero or carbon neutrality, with many technology companies setting 2030 as their target year for achieving these sustainability goals.[114] Corporations have a mixed record on satisfying these climate commitments,[115] however, and research has produced mixed results on whether these

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[112] *See* INT'L ORG. FOR STANDARDIZATION & INT'L ELECTROTECHNICAL COMM'N [ISO/IEC], INFORMATION TECHNOLOGY — ARTIFICIAL INTELLIGENCE — ENVIRONMENTAL SUSTAINABILITY ASPECTS OF AI SYSTEMS, ISO/IEC TR 20226:2025 (1st ed. July 2025), https://www.iso.org/standard/86177.html [https://perma.cc/X94B-RV7M].

[113] *See, e.g.*, Michael P. Vandenbergh, Jim Rossi & Ian Faucher, *The Gap-Filling Role of Private Environmental Governance,* 38 VA. ENV'T L. J. 1 (2020) (identifying the potential role of private environmental governance in motivating a public utility to reduce carbon emissions); CLIMATE IMPACT PARTNERS, QUIET CLIMATE ACTION 14, 17 (Sept. 2024) [hereinafter QUIET CLIMATE ACTION].

[114] QUIET CLIMATE ACTION, *supra* note 113, at 14.

[115] *See* THOMAS DAY ET AL., CORPORATE CLIMATE RESPONSIBILITY MONITOR 2024, at 5 (2024) (critiquing private commitments based on research concluding that even if 51 major global firms fulfill their commitments, they will reduce emissions by only 30 percent rather than the claimed 100 percent).

initiatives are responsible for significant emissions reductions and likely to experience similar results from AI initiatives in the future. To explore this issue, we present both an optimistic case and a pessimistic case for private governance initiatives.

A growing concern stems from our observation that employee recruitment and retention have historically been important drivers of corporate commitments to adopt sustainability goals and reduce their carbon emissions,[116] but the adoption of generative AI tools may be reducing demand for software engineers and other specialized technological professionals. [117] As demand for specialized technological expertise drops and unemployment rises, the balance of power between employees and employers may shift, causing employee concerns regarding sustainability to have a declining influence over their employers. Indeed, recent years have seen some backsliding among large corporate employers on sustainability and other employee-supported issues, although it is hard to ascribe a causal relationship to employee influence, given other evolving factors.[118] If this

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[116] *See* MICHAEL P. VANDENBERGH & JONATHAN M. GILLIGAN, BEYOND POLITICS: THE PRIVATE GOVERNANCE RESPONSE TO CLIMATE CHANGE 185, 292-92 (2017) (reporting that at technology companies such as Microsoft, environmental sustainability programs improve the company's ability to recruit and retain top talent).

[117] Natasha Singer, *Goodbye, $165,000 Tech Jobs. Student Coders Seek Work at Chipotle*, N.Y. TIMES (Aug. 10, 2025), https://www.nytimes.com/2025/08/10/technology/coding-ai-jobs-students.html [https://perma.cc/472S-ZSMM] (reporting that with the rise of AI coding tools, unemployment has grown rapidly among recently graduated computer science and software engineering majors from top universities, with recent grads facing twice the unemployment rate of art history majors). *See also* Jessica Dickler, *College Majors with the Best and Worst job Prospects—Art History Beats Finance*, CNBC (May 16, 2025), https://www.cnbc.com/2025/05/16/college-majors-with-the-best-and-worst-employment-prospects.html [https://perma.cc/2VA2-5XLW] (noting unemployment rates for recent graduates across various majors); *Why Pushing STEM Majors is Turning Out to be a Terrible Investment*, L.A. TIMES (Jan. 9, 2024), https://www.latimes.com/opinion/story/2024-01-09/science-jobs-technology-stem-majors [https://perma.cc/BY5L-YQS2] (discussing same).

[118] *See* Sunny Grimm, *Google Quietly Removes Net-zero Carbon Goal from Website Amid Rapid Power-hungry AI Data Center Buildout*, TOM'S HARDWARE (Sept. 5, 2025), https://www.tomshardware.com/tech-industry/google-quietly-removes-net-zero-carbon-goal-from-website-amid-rapid-power-hungry-ai-data-center-buildout-industry-first-sustainability-pledge-moved-to-background-amidst-ai-energy-crisis [https://perma.cc/BW8T-FWHV]; *Wall Street Wants You to*

proves true, declining employee power to turn down or leave employment at high-polluting firms could reduce corporate interest in private environmental governance.

The Optimistic Case. The optimistic case begins by identifying the extent of corporate emission reduction commitments and then examining likely compliance rates and effects on AI-driven data center emissions. However, identifying the companies that will create the most AI-related electricity demand and carbon emissions is difficult because corporate reporting is often incomplete or otherwise not transparent.[119] Some entities "cherry-pick" numbers to evade public scrutiny, which creates difficulties in assessing the true efficacy of corporate sustainability claims.[120] Nevertheless, a non-exhaustive list of the major AI companies with significant or growing energy footprints, includes Amazon, Meta, Microsoft, NVIDIA, Adobe, Palantir Technologies, Alphabet (Google), Intel, CoreWeave, OpenAI, Anthropic, and Tesla.[121]

Many of the companies that are making major investments in AI are leaders within the technology sector for climate-related commitments.[122] The analysis by Morgan Stanley discussed above suggests that several major AI firms have not backed off on their commitments following the 2024 election, although they may need to satisfy them now

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Know Profit Comes Before Net Zero, BLOOMBERG (Sept. 18, 2024), https://www.bloomberg.com/news/articles/2024-09-18/wall-street-wants-you-to-know-profit-comes-before-net-zero [https://perma.cc/2T9B-FC44]; Emma Goldberg, Aaron Krolik & Lily Boyce, *How Corporate America is Retreating from D.E.I.*, N.Y. TIMES (Mar. 13, 2025), https://www.nytimes.com/interactive/2025/03/13/business/corporate-america-dei-policy-shifts.html [https://perma.cc/2TBL-KG9S]; Wyatte Grantham-Philips & Haleluya Hadero, *Tractor Supply is Ending DEI and Climate Efforts After Conservative Backlash Online*, AP NEWS (June 28, 2024), https://apnews.com/article/tractor-supply-ends-dei-climate-goals-ab9e570d39095de6bead7fbfe76a6edc [https://perma.cc/E66F-JNFD]; Hanaa' Tameez, *From Reckoning to Retreat: Journalism's DEI Efforts are in Decline*, NIEMAN LAB (Sept. 11, 2025), https://www.niemanlab.org/2025/09/from-reckoning-to-retreat-newsrooms-dei-efforts-are-in-decline/ [https://perma.cc/HK3S-KB3P].

119 *See* Jonathan Gilligan, *Private Sector Wedge in Climate Mitigation Public Seminar* (Dec. 2, 2022).

120 *See id*.

121 *See* GREENING DIGITAL COMPANIES, *supra* note 66, at 25.

122 *See id.* at 34.

with complex mixes of clean power, offsets, and carbon capture and storage.[123] In September 2025, Google deleted its net-zero pledge from its sustainability website.[124] A review of corporate commitments by the leading AI firms conducted for this paper is included in Table 1.

Table 1. Climate Commitments by Leading AI Firms

Data Center Operators/Tech Firms	Climate Commitments
Google	Net-zero emissions across all operations and value chain by 2030 (until 2025).[125] Reducing 50% of combined Scope 1, 2 (market-based), and 3 absolute emissions (compared to their 2019 base year) by 2030.[126]
Amazon Web Services	Net-zero emissions by 2040, having matched 100% of their energy consumed with renewables by 2023.[127]

[123] *See* Kearney, *supra* note 88.

[124] Grimm, *supra* note 118. The effects of recent federal climate change regulatory retrenchment on corporate climate pledges are unclear. For instance, although Microsoft does not appear to have taken final definite action, it has recently been seriously considering its pledge to match all electricity use with renewables on an hour-to-hour basis by 2030. *See* Alastair Marsh & Brody Ford, *Microsoft in Talks to Ax Energy Pledge Amid Data Center Boom,* BLOOMBERG (May 6, 2026), https://www.bloomberg.com/news/articles/2026-05-06/microsoft-clean-power-target-on-chopping-block-over-data-center-boom [https://perma.cc/R2S2-ZMGH] (reporting that "[t]he company is weighing whether to delay — or even abandon altogether — its 2030 target of matching 100% of its hourly electricity use with renewable energy purchases, according to the people, who requested anonymity to discuss a private matter").

[125] *See id.*; *see also* Alexa St. John, *Google Falling Short of Important Climate Target, Cites Electricity Needs of AI*, AP NEWS (July 2, 2024), https://apnews.com/article/climate-google-environmental-report-greenhouse-gases-emissions-3ccf95b9125831d66e676e811ece8a18 [https://perma.cc/3DRT-ZHFY] (discussing previous climate commitment).

[126] GOOGLE, ENVIRONMENTAL REPORT 2025, at 83–85.

[127] *AWS Sustainability*, AMAZON WEB SERVS., https://aws.amazon.com/-sustainability/ [https://perma.cc/7EE6-B78R] (last visited Mar. 8, 2026); *Amazon Meets 100% Renewable Energy Goal 7 Years Early*, AMAZON NEWS (July 10, 2024), https://www.aboutamazon.com/news/sustainability/amazon-renewable-energy-goal.

Meta	Net-zero emissions across its value chain in 2030, with a reduction target in line with the Paris Climate Agreement.[128]
Microsoft	"By 2030, we'll be carbon negative. By 2050, we'll remove our historical emissions since our founding in 1975" (goals recently changed to a "moonshot").[129]
NVIDIA[130]	Scope 1 and 2: By the end of FY25, and annually thereafter, NVIDIA expects to achieve and maintain 100% renewable electricity for offices and data centers under their operational control.[131] Scope 3: By the end of FY26, NVIDIA expects to engage manufacturing suppliers comprising at least 67% of its Scope 3 category 1 greenhouse gas (GHG) emissions, with the goal of effecting supplier adoption of science-based targets.[132]
Adobe	Net zero by 2050; 42% reduction in absolute Scope 1 and 2 GHG emissions by FY 2030 from a FY 2022 base year; 52% reduction in Scope 3 GHG emissions per USD value added by FY 2030 from a FY 2022 base year.[133]

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[128] *Climate*, META SUSTAINABILITY, https://sustainability.atmeta.com/climate/ [https://perma.cc/9QFQ-KEZ Q] (last visited Mar. 8, 2026).

[129] Heather Clancy, *Microsoft Pivots as AI Makes Emissions Cuts Harder*, TRELLIS (Feb. 13, 2025), https://trellis.net/article/microsoft-pivots-climate-investments/ [https://perma.cc/9NTL-FAC8]. Microsoft's emissions have increased 30% since they announced their net-zero by 2030 goal, and Chief Sustainability Officer Emily Nakagawa said, "In 2020, Microsoft leaders referred to our sustainability goals as a 'moonshot,' and nearly five years later, we have had to acknowledge that the moon has gotten further away." Melanie Nakagawa, *Progress on the Road to 2030*, MICROSOFT: BLOGS (Feb. 13, 2025), https://blogs.microsoft.com/on-the-issues/2025/02/13/progress-on-the-road-to-2030/ [https://perma.cc/AQD6-RDSN].

[130] NVIDIA's GPUs underpin much of the AI boom, but as a component supplier, its own operational emissions (Scope 1 and 2) are relatively modest compared to the indirect emissions at large data centers using NVIDIA hardware. NVIDIA's focus is primarily on ensuring its direct operations are powered by renewables and pushing suppliers to reduce emissions. NVIDIA, SUSTAINABILITY REPORT FISCAL YEAR 2025, at 7, 29 (2025) [hereinafter NVIDIA FY25 SUSTAINABILITY REPORT], https://images.nvidia.com/aem-dam/Solutions/documents/NVIDIA-Sustainability-Report-Fiscal-Year-2025.pdf [https://perma.cc/DG8R-NGRC]; Heather Clancy, *NVIDIA's Top ESG Priority: Slash the Power Gobbled by Its AI Chips*, TRELLIS (Oct. 1, 2025), https://trellis.net/article/nvidia-top-esg-priority-slash-power-gobbled-by-ai-chips/ [https://perma.cc/7MP6-LV7A].

[131] NVIDIA has reported achieving this goal ahead of schedule. NVIDIA FY25 SUSTAINABILITY REPORT, *supra* note 130, at 12.

[132] This goal has also been reportedly achieved ahead of schedule. *Id.*

[133] *Adobe Sustainability Policy* (updated Oct. 2025), https://www.adobe.com/corporate-responsibility/sustainability/corporate-policy.html [https://perma.cc/5ECQ-JXR9].

Palantir Technologies	Achieved carbon neutrality in 2023.[134]
Intel	Achieve a 10% reduction in Intel's absolute Scope 1 and 2 GHG emissions by 2030 and reach net-zero Scope 1 and 2 GHG emissions by 2040.[135] Scope 3: Net zero by 2050.[136]
CoreWeave	No readily identifiable concrete commitment
OpenAI	No readily identifiable concrete commitment[137]
Anthropic	No readily identifiable concrete commitment[138]
Tesla	No readily identifiable concrete commitment
xAI	No readily identifiable concrete commitment[139]

We were unable to identify an analysis of the aggregate emissions reductions that would arise from the fulfillment of these commitments. However, if the firms adhere to their commitments, despite pressure from increased electricity demand, the impacts will

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134 *See* PALANTIR TECHS., 2023 CARBON EMISSIONS REPORT 2 (2024), https://www.-palantir.com/assets/xrfr7uokpv1b/6DpfIkl0r7IaAHJ97kadA9/58b30f26cdb627060a96070d788f5c34/-FINAL-_2023_Carbon_Emissions_Report.pdf [https://perma.cc/JP33-FRVF].

135 *Intel and the Environment*, INTEL, https://www.intel.com/content/www/us/en/environment/intel-and-the-environment.html [https://perma.cc/2ESX-6M8C] (last visited Mar. 8, 2026).

136 INTEL, 2024-25 CLIMATE TRANSITION ACTION PLAN (2025), https://www.intel.com/-content/dam/www/central-libraries/us/en/documents/2026-01/climate-transition-action-plan.pdf [https://perma.cc/RD7F-L8H3].

137 OpenAI uses Microsoft Azure as its primary cloud backend for training and serving (meaning some emissions are effectively captured under Microsoft's targets). *See* FTC OFF. POL'Y PLAN., PARTNERSHIPS BETWEEN CLOUD SERVICE PROVIDERS AND AI DEVELOPERS (2025), at 20 n.149, https://www.ftc.gov/system/files/ftc_gov/pdf/p246201_aipartnerships6breport_redacted.pdf [https://perma.cc/5G6L-85GA] (finding that OpenAI has used Microsoft Azure as its "exclusive cloud provider" since 2019, reaffirmed in 2023).

138 Like OpenAI, Anthropic predominantly uses external cloud providers (reports indicate AWS). *See Amazon and Anthropic Deepen Strategic Collaboration*, AMAZON NEWS (Nov. 22, 2024), https://www.aboutamazon.com/news/aws/amazon-invests-additional-4-billion-anthropic-ai [https://perma.cc/EA77-5ZF9]; *Powering the Next Generation of AI Development with AWS*, ANTHROPIC (Nov. 22, 2024), https://www.anthropic.com/news/anthropic-amazon-trainium [https://perma.cc/773N-J32P] (both confirming AWS as Anthropic's primary cloud and training partner).

139 xAI has installed dozens of gas turbines to power its Memphis TN data center. *See* Jennifer Ugwa, *In South Memphis, Elon Musk's Colossus Operated Gas Turbines without Appropriate Permits, Residents and Activists Claim,* INSIDE CLIMATE NEWS (July 17, 2025), https://insideclimatenews.org/news/17072025/elon-musk-xai-data-center-gas-turbines-memphis/ [https://perma.cc/ZA7J-XS3D].

likely be in the tens of millions of tons.[140] Analysts already note that some firms are likely to miss their commitments, to scale back their commitments, or to meet their commitment with low quality offsets, so these companies' actual emission reductions are likely to be less than their commitment amounts.[141] Even if only half of these commitments are achieved, they will amount to substantial reductions from the government regulation-only scenario.[142]

In addition, research suggests that firms participate in private environmental governance activities, such as climate commitments, for several reasons. Namely, firms might participate for employee recruitment and retention purposes, as well as efforts to identify efficiencies, supply chain pressures from companies that have incentives to reduce emissions, retail customer pressures, investor, lender, and insurer pressures, local community pressures, and anticipation of future regulations.[143] This complex mix of motivations, combined with the inability to enforce companies' previous commitments, makes accountability difficult to assess.

Climate commitments, the recent efforts by AI firms, such as Amazon, to re-start nuclear plants or otherwise identify clean energy sources for data centers,[144] and the various factors leading to participation in private climate governance suggest that

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140 *See* U.S. ENV'T PROT. AGENCY, INVENTORY OF U.S EMISSIONS AND SINKS 1990-2022 (2024).

141 *See* THOMAS DAY ET AL., *supra* note 115, at 5 (critiquing private commitments based on research that concluded that even if 51 major global firms fulfill their commitments, they will reduce emissions by 30 percent, rather than the claimed 100 percent).

142 *Cf.* INT'L ENERGY AGENCY, *supra* note 20, at 18 (modeling data center emissions in the "base case" scenario—i.e., under current regulatory conditions); Wincele, *supra* note 97 (noting that the largest AI data center buildout is occurring in states without meaningful emissions requirements).

143 *See* VANDENBERGH & GILLIGAN, *supra* note 116; Michael P. Vandenbergh, *Sidebar: The Drivers of Corporate Climate Mitigation*, 35 ENV'T. F. 29 (2018).

144 Microsoft has pursued the nuclear option as well. *See Three Mile Island Nuclear Plant to Reopen, Sell Power to Microsoft,* CBS NEWS (Sept. 21, 2024, at 21:54 ET), https://www.cbsnews.com/news/three-mile-island-microsoft-nuclear-power/ [https://perma.cc/42S9-D484].

corporations feel substantial pressure regarding their AI-related emissions. The new U.S. administration's anti-climate mitigation initiatives threaten to undercut corporate decarbonization initiatives. Although the pressure from employees, retail consumers, foreign supply chain buyers, and many investors, lenders, and insurers may be reduced, it is unlikely for such pressure to completely disappear. This dynamic suggests that advocacy groups and states advocating for climate change mitigation can be expected to increase their focus on inducing firms to make and comply with corporate climate commitments. Ironically, AI may enable advocacy groups to identify the carbon emissions arising from data centers and the AI firms that use them more quickly and comprehensively, enabling themselves to better identify greenwashing by firms with carbon commitments and pressure corporations to reduce carbon emissions from AI-serving power generation.

The Pessimistic Case. The pessimistic case begins with a much more negative view of corporate climate pledges than the optimistic case. While the optimistic case assumes that corporate climate pledges influence firm behavior, a substantial amount of literature casts doubt on the effects of these pledges.[145] Transparency is uneven and largely self-policed, and a survey of tech firms found many were unaware of new energy reporting obligations or unprepared to comply with them.[146] Moreover, such literature provides examples of corporations that have made climate commitments, but ignore the finance and data/AI sector leaders who are calling for the abandoning of climate goals to pursue greater

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[145] *See, e.g.*, FREDERIC HANS ET AL., CORPORATE CLIMATE RESPONSIBILITY MONITOR 2025, at 51 (2025), https://newclimate.org/sites/default/files/2025-09/newclimate-publication-ccrm2025-full-report-jul25.pdf [https://perma.cc/M4QY-4DLQ].

[146] *See* Kai Ebert et al., *AI, Climate, and Regulation: From Data Centers to the AI Act,* ARVIX (Oct. 9, 2024), https://arxiv.org/html/2410.06681v1.

energy consumption for AI (e.g., former Google CEO Eric Schmidt).[147] Amazon's own employees have condemned the firm's climate pledges as empty greenwashing.[148] Additionally, with Google and Microsoft shifting from describing their net-zero ambitions as commitments to titling them "moon-shots"—emphasizing how risky and unlikely they are—a growing number of AI and technology firms may be under pressure to reduce or abandon their commitments.[149]

Engineering research suggests many of the commitments listed in Table 1 are exaggerated and unlikely to produce significant emissions reductions. For instance, Amazon's claim to match its energy consumption with renewable generation was found to mostly displace some forms of renewable energy with other forms, thus having a limited impact on displacing fossil-fuel consumption or reducing emissions.[150] Additionally, despite claiming initiatives to reduce greenhouse gas emissions, Google's emissions actually rose 13% in 2023.[151] More broadly, there is evidence that companies with sustainability commitments and high scores for their environmental, social, and

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147 David Gelles, *The A.I Power Grab*, N.Y. TIMES (Oct. 22, 2024), https://www.nytimes.com/-2024/10/22/climate/ai-big-tech-emissions.html [https://perma.cc/67RJ-WLD3].

148 *See* AMAZON EMPS. FOR CLIMATE JUST., BURNS TRUST: THE AMAZON UNSUSTAINABILITY REPORT 14–16 (2024), https://static1.squarespace.com/static/65681f099d7c3d48feb86a5f/t/6721c4047213ea343e50536f/1730266118471/unsustainability-report-2.pdf [https://perma.cc/Z8C2-G4LA].

149 *See* Grimm, *supra* note 118 (reporting that Google has removed its net-zero pledge from its corporate sustainability page and altered a related net-zero pledge on its data center sustainability page to say that net-zero is a "moonshot" aspiration rather than a firm commitment). *See also* Clancy, *supra* note 129 (quoting Microsoft Chief Sustainability Officer Melanie Nagakawa saying that, "In 2020, Microsoft leaders referred to our sustainability goals as a 'moonshot,' and nearly five years later, we have had to acknowledge that the moon has gotten further away").

150 James Temple, *Google, Amazon, and the Problem with Big Tech's Climate Claims*, MIT TECH. REV. (July 17, 2024), https://www.technologyreview.com/2024/07/17/1095019/google-amazon-and-the-problem-with-big-techs-climate-claims/ [https://perma.cc/6GNZ-F8RE]; *see* Qingyu Xu et al., *System-Level Impacts of Voluntary Carbon-Free Energy Procurement Strategies*, 8 JOULE, 374, 374–400 (2024) (establishing that Amazon uses an emissions matching strategy, which often results in new procurement of low-carbon energy replacing old procurement of low-carbon energy).

151 *See* Temple, *supra* note 150.

governance (ESG) policies pollute as much as their competitors with low scores.[152] Rising emissions with double-digit growth rates are not consistent with a credible pledge to bring them to zero quickly. In short, the pessimistic case requires a fuller discussion of corporate climate pledges—one that treats calls to reject energy and carbon constraints on equal footing with celebrations of corporate climate commitments and engagement with engineering analyses questioning whether the offsets included in those pledges reduce emissions on net.

A successful assessment of the AI sector's future emissions would focus on the previous over-optimism surrounding the economic growth potential of the tech sector during the dot-com boom and bust (which included the collapse of various high-flying start-ups, as well as large established firms, such as AOL and Yahoo), as well as macroeconomic growth. If we adopt a realistic view of the tech sector's future and the national and global economy, emissions forecasts will decline substantially. Part of the research agenda would involve generating more realistic—and thus pessimistic—economic growth forecasts.

Research Agenda. The challenge for rational policymaking is to identify public and private governance responses that enable AI development while preventing the growth of AI-generated electricity demand that undermines the ability to meet environmental targets. We suggest research initiatives on several topics to identify feasible options:

Applying Rigorous Engineering Assessments to the Emissions Reductions Produced by Corporate Pledges. It is difficult to assess private governance initiatives

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[152] Steve Johnson, *Companies with Good ESG Scores Pollute as Much as Low-Rated Rivals*, FIN. TIMES (Jul. 31, 2023), https://www.ft.com/content/b9582d62-cc6f-4b76-b0f9-5b37cf15dce4 [https://perma.cc/AB8R-X6UR].

without credible, independent assessments of the emissions reductions they claim. To close this knowledge gap, it is necessary to apply recognized, rigorous engineering methods to assess the actual emissions of data-intensive corporations and report the annual emissions reductions achieved under their sustainability policies.

Stimulating and Reinforcing Private Climate Commitments. If the best estimates by energy analysts are correct, private net zero and other commitments will play an important role in enabling carbon emissions targets to be met, despite the increases in AI electricity demand. However, many aspects of these commitments are not fully understood and can easily fall prey to ideological biases regarding their effectiveness, causing a spillover effect on public governance responses. Important questions include: (1) What combination of factors (e.g., employee recruitment and retention; retail customer demand; supply chain pressure; efficiency gains; investor, lender and insurer pressure; and the anticipation of future regulation) motivates companies to adopt climate commitments?; (2) How likely are these climate commitments to be fulfilled?; (3) To what extent do advocacy groups target firms that make commitments regarding greenwashing versus those that hide in the weeds?; (4) Do advocacy group and academic critiques of firms that make commitments but fall short increase or decrease a firm's motivations to make commitments, and does descriptive norm theory suggest a different approach?; (5) What is the effect of private sector commitments on the adoption of international, national, and subnational governmental requirements regarding carbon emissions?; and (6) How can AI be used to assess and publicize compliance with carbon commitments?

Avoiding a Race to the Bottom. A potential reason to include private governance interventions in the response to data center load growth is that absent private motivations,

the lack of federal emissions reductions requirements, combined with the existence of carbon safe haven states, could lead new data centers and power plants to be placed in states without regulatory requirements or substantial tort risks. In theory, states could face race-to-the-bottom incentives that attract these businesses, and these incentives could reinforce a resistance to adoption of carbon emissions targets or statutory liability, along with other climate governance options. The legal literature on the race-to-the-bottom effect is unsettled, however, and research that accounts for the attributes of AI, data centers, and the power plants that serve them as well as the public and private governance options could shed light on the optimal mix of governance responses.

Understanding the Influence of Employees. Employee pressure has been a powerful force in persuading corporations to embrace sustainability goals. As AI displaces workers, employers have greater power to resist employee pressures and abandon private environmental governance. Research should examine the mix of worker protections, such as the ability to express views in the workplace, on firm pressure to adopt corporate strategies for lower costs, greater efficiency, and reduced environmental impacts.

Analyzing the Role of Climate Activists in Private Governance. Private governance analyses have largely focused on cooperative measures that work with businesses to reduce emissions and the drivers of corporate motivations to participate in such initiatives. Additional research should investigate the role of advocacy group reputation campaigns, consumer boycotts, and other efforts directed toward specific firms or entire economic sectors. There is anecdotal evidence that attacks on corporate reputations have powerful effects, but further work must be done on the effects of these attacks politically from the

left and the right.[153] It is striking that there has been a sharp and bipartisan decline in trust among both large corporations and the banking and finance sector in recent years. More than two thirds of the U.S. public, including roughly equal numbers of Republicans and Democrats, agree that large corporations and the banking and finance sectors are bad for America.[154] Broad mistrust of large corporations and financial institutions suggests fertile ground for research on the effects of these negative perceptions on private governance initiatives. Thus, it may be valuable to study the successes and failures of the Occupy Wall Street and anti-World Trade Organization movements on the left, as well as those of the more recent anti-ESG and anti-Diversity, Equity, and Inclusion campaigns on the right.[155]

V. CONCLUSION

The expected electricity demand from AI-driven growth is highly uncertain given fast-paced technological improvements and indeterminacy about the amount and type of

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153 *See, e.g.*, Hanna Ziady, *Bud Light Boycott Likely Cost Anheuser-Busch InBev Over $1 Billion in Lost Sales,* CNN BUS. (Feb. 29, 2024), https://www.cnn.com/2024/02/29/business/bud-light-boycott-ab-inbev-sales/index.html; Jessica Guynn, *Battered by Boycott and Backlash, Target Will Not Sell Pride Collection in All Stores*, USA TODAY (May 9, 2024), https://www.usatoday.com/story/money/2024/05/09/target-pride-month-collection-backlash-fewer-stores/73636054007/ [https://perma.cc/B3BZ-MBLX]; Brooke Masters & Patrick Temple-West, *The Real Impact of the ESG Backlash*, FIN. TIMES (Dec. 4, 2023), https://www.ft.com/content/a76c7feb-7fa5-43d6-8e20-b4e4967991e7 [https://perma.cc/8QHU-H4HJ]; Greg Ritchie & Jacqueline Poh, *A $1.5 Trillion Loan Market Feels the Fallout of Anti-ESG Forces*, BLOOMBERG L. (Nov. 17, 2023), https://www.bloomberg.com/news/articles/2023-11-17/a-1-5-trillion-loan-market-feels-the-fallout-of-anti-esg-forces [https://perma.cc/96T4-XUBC]; Grantham-Philips & Hadero*, supra* note 118*;* Emma Goldberg, *How Companies Are Navigating the D.E.I. Backlash*, N.Y. TIMES (Feb. 10, 2025), https://www.nytimes.com/2025/02/10/us/politics/dei-corporate-america.html [https://perma.cc/3D7N-ZXUX].

154 *From Businesses and Banks to Colleges and Churches: Americans' Views of U.S. Institutions*, PEW RSCH. CTR., (Feb. 1, 2024), https://www.pewresearch.org/politics/2024/02/01/from-businesses-and-banks-to-colleges-and-churches-americans-views-of-u-s-institutions/ [https://perma.cc/X9QH-LPKC].

155 *See* James A. Anderson, *Some Say Occupy Wall Street Did Nothing. It Changed Us More Than We Think*, TIME (Nov. 15, 2021), https://time.com/6117696/occupy-wall-street-10-years-later/ [https://perma.cc/V6AW-GH42] (analyzing the effects of Occupy Wall Street); Micah L. Sifry, *Occupy Wall Street at 10: What It Taught Us, and Why It Mattered*, NEW REPUBLIC (Sept. 17, 2021), https://newrepublic.com/article/163680/occupy-wall-street-10-year-anniversary-lessons [https://perma.cc/CU7W-HFAF] (same).

demand from retail and corporate users. On one hand, companies recently scaled back investments in new computing capacity.[156] On the other hand, significant increased electricity demand and environmental impacts from growing AI-driven data center use could pose a fundamental challenge to the United States' ability to both provide low-cost electricity and meet environmental goals. The projected growth could lock in capital expenditures for new power plants and transmission and may lead to decades of elevated emissions. National and subnational political turbulence in the United States suggests that major government incentives are unlikely to address this challenge in the near term, even though waiting years for government action will lead to capital expenditures locking in future demand and emissions.

This Article provided a new perspective on the debate about AI-driven electricity demand by examining the history of over-predictions, identifying viable initiatives that may be able to shift the amount or timing of demand, and avoiding the tendency to propose federal or state regulatory restrictions on AI use or carbon emissions that are infeasible for the foreseeable future. As to household behavior, this Article identified examples of successful energy and environmental disclosure programs and presented exploratory new empirical data to explain how disclosing electricity use and carbon emissions may shift individual and corporate behavior. As for AI firms, this Article identified the strengths and

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[156] *See, e.g.*, Georgia Butler, *Microsoft Cancels 200 mW of AI Data Center Leases*, https://www.datacenterdynamics.com/en/news/microsoft-cancels-200mw-of-ai-data-center-leases-report/ [https://perma.cc/JC3S-29VP] (indicating that Microsoft has withdrawn from data center leasing agreements worth 200 mW); Matthew Gooding, *Microsoft Paused Construction on Part of Data Center Site in Mount Pleasant, Wisconsin*, DATA CTR. DYNAMICS (Jan. 3, 2025), https://www.datacenterdynamics.com/en/news/microsoft-pauses-construction-on-part-of-data-center-site-in-mount-pleasant-wisconsin/ [https://perma.cc/4QP5-FAYY] (reporting that Microsoft paused construction of part of a data center in Mount Pleasant, Wisconsin, signaling reduced Microsoft need for direct data center services).

weaknesses of corporate climate commitments and discussed ways to reduce the risk of greenwashing while enhancing these commitments' gap-filling role. Additional research on each of these topics will generate insights supporting efforts to enable AI to reach its potential while reducing emergent energy and environmental challenges.

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